

Strict twice partitions

Phase-resolved asymptotics for A271619,
and nested partitions with distinct block lengths (A358836)

Part I: Report 182 (author line “A rigorous asymptotic and reproducibility report”)

Part II: Research report 181

3 October 2026; merged 5 October 2026 (batch 100)

Abstract

Let $a(n) = [q^n] \prod_{k \geq 1} (1 + p(k)q^k)$, where $p(k)$ is the ordinary partition number. These are twice-partitioned numbers with a strict outer partition, recorded as OEIS A271619. We prove a phase-resolved expansion to every fixed relative order. Its two effective upper-edge charges exchange dominance near a fractional triangular phase $9/16$, with a positive $(\log M)/\sqrt{M}$ shift and an $M^{-1/2}$ transition width. A persistent independent low-hole cloud supplies the leading multiplier $C_0 = \prod_{k \geq 1} (1 + 1/p(k))$ and higher finite-moment corrections. The proof combines an exact unique filling map, quantitative per-hook localization, controlled exponential moments, shifted-symmetric quasimodularity, and coefficient extraction with explicit minor-arc control. A smooth finite-charge carrier yields integer inverse enclosures with two ceilings; a positive- L^1 argument also gives a uniform joint charge/hole approximation and actual-cardinality moments. Classical partition, Fermi, quasimodular, and asymptotic methods are explicitly credited.

Finite-index warning. This is an eventual asymptotic theorem, not a certified practical approximation at moderate indices. At $n = 5000$, a floating evaluation of the leading two-charge carrier divided by the exact coefficient is approximately 0.00177838; including the first correction gives approximately 0.00486243. An ordinary single-saddle Gaussian happens to give approximately 1.00047777 there. These observations do not establish its eventual validity. The asymptotic on-set is not made effective. Even at $M = 1000$, the smooth equal-weight phase is approximately 1.02188, outside the physical cell $[0, 1]$. All such floating diagnostics are separated from the exact computations and rigorous constant intervals.

Scope and provenance. The sequence and its defining product are prior work of Gus Wiseman (2016) in the inspected OEIS entry [3]. The substantive result here is the target-specific relative expansion and its controlled errors. Our bounded source screen found no directly covering theorem; it is not a global priority certification, and no claim of being first is made. This report and its accompanying code have not been submitted externally.

Editorial note ([write], 5 October 2026, batch 100). This report is built from two manuscripts of one external research session: Reports 182 and 181 of the session bundle of Reports 1–243, which arrived in ProveIt commit 60f54ea06 and were placed together, as cluster DPP of batch 100 of the incoming-reports intake, in 36571ae0e. Report 182 (Part I, the base; its abstract, finite-index warning and scope paragraph stand above) treats OEIS A271619, $a(n) = [q^n] \prod_{k \geq 1} (1 + p(k)q^k)$: collections of integer partitions whose *sizes* are distinct. Report 181 (Part II) treats OEIS A358836, $a_n = [q^n] \prod_{k \geq 1} (1 + q^k/(q; q)_k)$: collections of integer partitions whose *lengths* (numbers of parts)

are distinct. Both are dated 3 October 2026. Report 182’s package adapts Report 181’s build scaffold, as its source audit says, but neither manuscript uses the other’s mathematics: they prove no common theorem and share no proof. Each Part keeps its own definitions, hypotheses and notation. Section 2 tabulates the letters the Parts use differently, above all b , which is the constant $\pi\sqrt{2/3}$ in Part I and the variance $L''(t_n)$ in Part II.

Status. Part I’s author line, “A rigorous asymptotic and reproducibility report”, is that manuscript’s description of itself, not an assessment by anyone else. Neither manuscript has been refereed; the reviews named in Part I’s review record (Section 14) belong to the delivering session; no statement of this report has been checked by a proof assistant. Part I’s main theorem is an *eventual* result only: at $n = 5000$ its leading formula is about 562 times too small, and the first-corrected one about 206 times (its finite-index warning above and Section 13.3).

Text added in the write is marked [write]. Sections 1–3 are front matter written for this report, Section 15 and Section 27 close each Part with its further questions (the manuscripts’ own, and the claims the write found stated without proof), and Appendix A records the provenance, the numbering crosswalk and every editorial change.

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1 How the two Parts fit together

[write] Both sequences count *twice partitions with a strict outer level*: an object is a finite collection of ordinary partitions (the blocks), and no two blocks share the value of one statistic. Both generating functions are decorated distinct-part products $\prod_{k \geq 1} (1 + w_k q^k)$.

	Part I: A271619 (Report 182)	Part II: A358836 (Report 181)
distinct statistic	block size (weight)	block length (number of parts)
product	$\prod_k (1 + p(k)q^k)$, (1)	$\prod_k (1 + q^k P_k(q))$, $P_k = 1/(q; q)_k$, (84)
decoration w_k	$p(k)$: fixed, $\log p(k) \sim \pi \sqrt{2k/3}$	$1/(q; q)_k \rightarrow P(q)$: depends on q ; at the saddle $\log P(e^{-t}) \sim A/t \asymp n^{1/4}$
first terms	1, 1, 2, 5, 8, 18, 34, 65, ...	1, 1, 2, 4, 8, 15, 28, 51, ...
leading growth	$\log a(n) \sim \frac{4\pi 2^{1/4}}{3\sqrt{3}} n^{3/4}$, constant 2.87598 (Proposition 5.2)	$\log a_n \sim \frac{4\pi}{3 \cdot 24^{1/4}} n^{3/4}$, constant 1.89250 (Theorem 21.1)
mechanism	two effective charges $M, M-1$ (a filled staircase plus an edge partition) exchanging dominance at the triangular phase 9/16, with a persistent independent low-hole cloud; a single central saddle provably fails (Section 6)	one exact saddle, valid at every angle (Lemma 19.1), after an exact reduction to Jacobi, partition and MacMahon products (Section 17)
outer count K_n	$M - B_n - \mathcal{H}_\infty $ in total variation, bounded fluctuations (Theorem 12.1)	Gaussian, $\text{Var } K_n \sim 1/(3t_n)$ (Theorem 23.1)

The two growth constants are computed from Proposition 5.2 and (117); both outer counts are of order $\sqrt{n}$ (Part I: $M \sim \sqrt{2n}$; Part II: $K_n/\sqrt{n} \rightarrow \sqrt{2/3}$).

- **Part I** (Sections 4–15): an expansion of $a(n)$ to every fixed relative order through a smooth two-charge carrier (Theorem 4.1), with the explicit first and second corrections (12), (72); the discrete inverse (Theorem 4.2); elementary growth and monotonicity (Section 5); the failure of a single central minor-arc bound (Section 6); the exact low-hole filling identity (Section 7); the tame edge expansion through Bloch–Okounkov brackets (Theorem 8.1); filled-block localization (Theorem 9.1); the crossover centre and the product normalization (Section 11); the joint law of the filled charge and the low holes (Theorem 12.1); and pure-integer enclosures of C_0 , μ_1 , μ_{count} and σ_{count}^2 (Section 13).
- **Part II** (Sections 16–27): the exact reduction (87) and the finite-end expansion (Proposition 17.2); the free energy to every fixed order (Theorem 18.1); an all-angle bound (Lemma 19.1) and every fixed Edgeworth order (Theorem 20.1); three power-log tiers and an elementary relative equivalent (Theorems 21.1 and 21.2); controlled inversion (Section 22); the Gaussian law of the number of blocks (Theorem 23.1); and the Gumbel law of the largest length with joint independence (Theorem 24.1).

What the write adds. Proofs of two steps that Part II states in outline: the sharpened mean (135) (after the proof of Theorem 23.1) and the marker-derivative bounds used in the proof of Theorem 24.1; a one-line justification of a sentence of Section 11; dated notes; and the lists of unproved claims in Sections 15 and 27. Nothing in either manuscript was found to be wrong. One wording is corrected in a note: Part II calls a $\exp(-ct^{-2\delta})$ bound “exponentially small” (proof of Theorem 20.1).

Relation to other reports in the repository. The neighbouring reports named here are in the directory

- **a291698-moving-fugacity-partitions** (labels **mfp:**) treats $[q^n] \prod_k (1 + n^\alpha q^k)$. Its exact Jacobi decomposition, its equation (11) in the section “Exact Jacobi decomposition and the finite-sector proof”, continues analytically to complex fugacity; taken at $u = e^{\mu(z)}$ its logarithm is Part II’s expansion (96) of $\log D(z)$. Part II derives it independently from the triple product and adds the dual-term bound on its moving complex disk (see the note after (96)). Its Section 10.3 asks for the regime $\log u \asymp n^{1/6}$ beyond polynomial fugacity. Part II’s effective fugacity has $\log u \asymp n^{1/4}$ and depends on q itself; Part II’s own source audit says that direct substitution into that report’s theorem is not justified, and its all-angle bound leaves no resonance sector at all. Part II is a neighbouring model, not an answer to that question.
- **a022629-distinct-partition-norms** (labels **dnp:**) treats $\prod_k (1 + k^\alpha q^k)^\mu$ and slot multiplicities. Its question 7 asks which smooth regularly varying weights preserve a single large crossing. The weight $p(k)$ of Part I grows like $e^{\pi\sqrt{2k/3}}$, is not regularly varying, and produces two competing charges instead of a single crossing: a different regime, not an answer. In the same batch 100, Report 93 of the bundle is being written into that report as its Section 22 (a critical theta window for a growing power): the same class of phenomenon as Section 6 here, near-modulus-one Fourier peaks that defeat a single Gaussian saddle, for a different product; no theorem is shared.
- **a126348-stable-hilbert-series** (labels **shs:**, new in batch 100, Report 84 of the bundle) treats $\prod_k (1 + q^k/(1 - q))$, whose fugacity $1/(1 - q)$ also depends on q ; like Part II it uses Jacobi’s triple product and the eta transformation. No theorem is shared.
- No other report, volume or formal development treats A271619, A358836 or their neighbour A261049 (searched 5 October 2026).

Formal status. No statement of this report is formalized. The classical identity both Parts start from, Jacobi’s triple product, is formalized in Lean as `Fabius.hasSum_jacobi_triple_product` in

`Analysis/FabiusFunction/Lean/FabiusFunction/JacobiTripleProduct.lean`:

$\sum_{k \in \mathbb{Z}} (-1)^k q^{k(k-1)/2} z^k = (z; q)_\infty (q/z; q)_\infty (q; q)_\infty$ for $|q| < 1$, $z \neq 0$. Part I’s normalization (29) is its instance $q = e^{-\tau}$, $z = -q^{1/2}$, and Part II’s identity (95) its instance $q = e^{-z}$, $z \mapsto -e^{-\mu(z)}$. Nothing beyond that classical identity is formalized.

2 Notation across the two Parts

[write] No symbol of either manuscript was renamed. Each Part uses its own manuscript’s letters, defined where they first occur. The following mean the same in both Parts: $A = \pi^2/6$; $\mathbb{E}$ (expectation) and Var ; $q = e^{-t}$ for a real or complex parameter t ; K_n , the number of outer blocks of a uniformly chosen object of size n ; $p(k)$, the partition function (Part II writes $P(q) = \sum_k p(k)q^k$); and Y , the logarithm of a threshold in the inverse problem. The letters in Table 1 mean different things in the two Parts; the right-hand column names the tempting false reading.

Table 1: Letters with different meanings in Parts I and II.

Letter	Part I (Report 182)	Part II (Report 181)	Not to be read as
b	the constant $b = \pi\sqrt{2/3}$ of (2), so that $\log p(k) \sim b\sqrt{k}$	the variance $b = \kappa_2 = L''(t_n) \sim 6A^2t_n^{-5}$ of (105); $b(s) = L_{tt}(\tau(s), s)$	Part II's b grows like $n^{5/4}$ and is not a constant;
a	$a(n)$, A271619; the Taylor increment $a(h)$ of Section 8; the constant $a = 21/5$ of Section 8.6	a_n , A358836; the constant $a = A^2/2 = \pi^4/72$ of (86)	Part I's b is not a variance $a(n)$ and a_n are different sequences; $21/5 \neq \pi^4/72$
B	corrections $B_1(c), B_2(c)$, (12), (72); normalization $B(N)$, (44); Bernoulli variables B_k (Section 6) and B_n (Section 12); $B_{m,r;L}$	block series $B_k(q) = q^k P_k(q)$, (84); the tier constant B_n of (114)	Part I's B_n is a random variable, Part II's B_n a number; neither B_k is the other
C	$C_0 = \prod_k (1 + 1/p(k))$; the growth constant C of Proposition 5.2; C_P of (77)	the tier constant C_n of (114); $c = 4s_*/3$ in Section 22	
D	tame cutoff $D = \lfloor m^\delta \rfloor$; $D(\varphi)$ of Section 11; inverse margin D_J	the Jacobi product $D(z)$, $D_s(z)$; $D_n = B_n - A/4$; the disk $\mathcal{D}_t$	
E	charge carriers $E_J(x)$, (7); Eisenstein series E_2, E_4, E_6 ; E_m^{tame}	Edgeworth corrections $E_h(t)$, in particular E_1, E_2 , (111), (112)	Part I's E_2 is an Eisenstein series or a carrier (Section 10.4 says which), Part II's E_2 the second Edgeworth correction
F	$F(t) = \log \prod_k (1 + p(k)e^{-tk})$, a logarithm	the generating function $F(q)$; its logarithm is $L(t)$	Part I's F corresponds to Part II's L
L	the low cutoff $L = \lfloor M^{17/20} \rfloor$	the free energy $L(t) = \log F(e^{-t})$, $L(z, s)$, L_2	
M	$M = M(n) = \lfloor (\sqrt{8n+1} - 1)/2 \rfloor$; M_N for size N	MacMahon's product $M(q)$; the random largest occupied length M_n	Part II's M_n is random, Part I's M_N a deterministic index
N	$N(Y) = \min\{n : \log a(n) \geq Y\}$; a total size N (Section 9); the energy N of Section 8	$N(x) = \max\{n : a_n \leq x\}$, $Y = \log x$; N_0	not one definition: applied to one sequence, Part II's rule at $x = e^Y$ gives Part I's $N(Y) - 1$, unless e^Y is a value of the sequence ($Y > 0$)
P	$P_m = \prod_{k \leq m} p(k)$, a number; $\mathcal{P}(q) = \prod_k (1 - q^k)^{-1}$	$P_k(q) = \prod_{j \leq k} (1 - q^j)^{-1}$ and $P(q)$, power series	Part II's $P(q)$ is Part I's $\mathcal{P}(q)$, not P_m
Q	shifted power sums $Q_j(\lambda)$, (26), in particular Q_2, Q_3 ; $Q(c) = V(c) - S(c)$; the filled charge Q_n	the inverse carrier $Q_2(t)$, (123)	the two Q_2 are unrelated
R	$R_{m,r}^{(D)}$; $R_r = r - 1/24$	the finite-end factor $R(z)$, $R_k(z)$; the Taylor remainder R_H	
T	triangular numbers $T_m = m(m+1)/2$; tails $T_s(N)$, (82)	$T_k(z) = \mu(z) - \mu_k(z)$; the tail sum $T_m(z, s)$, (138)	Part II's T_m is not triangular
Z	the normalization $Z(\tau)$, (29); $Z_m(s; D)$	$Z = \zeta(3)$; a standard normal Z_0	

Letter	Part I	Part II	Not to be read as
G, H, I	$G(d), G_\ell(d)$; the hole set $\mathcal{H}$ and the width H_t ; Bessel functions I_ν	the Gumbel variable G , geometric G_j, G_n ; the formal series $H(z)$ and the Edgeworth order H ; the interpolant I	
J	a fixed expansion order (never a charge)	a fixed truncation order; in (109)	
S, U, V, W	$S(c), S_D, S_t$, the outer set S ; Bernoulli U_h, V_h ; $U = Q_3 + 7/960$; $V(c)$; $W_m(n)$, $W(u)$	$U = t/\varepsilon$ in the proofs of Theorems 21.1 and 21.2	
t	the physical parameter, $t = f(M)/M \sim b/\sqrt{M}$ in Section 7.1; $X(t)$	the exact saddle $t = t_n \sim s_* n^{-1/4}$	
c, m	$c = r/m$, $c_m(x)$; m an effective filled-edge charge	c a small disk constant; m a threshold index, m_j multiplicities, $m_B(t)$	
μ	hole moments $\mu_j = \mathbb{E}h_\infty^j$; μ_{count}	$\mu(z) = \log P(e^{-z})$, $\mu_k(z)$	Part II's $\mu(t) \sim A/t$ is not a moment
κ, τ	κ positive constants; $\tau = \pi/\sqrt{6r}$	cumulants κ_j ; the complex saddle $\tau(s)$ and the root τ of $Q_2(\tau) = Y$	
ℓ, h	$\ell = L/m$ and a resonance index; the hole energy $h(\mathcal{H})$, h_∞ , half-integer positions h	$\ell = \log(t/2\pi)$, ℓ_n ; coefficients h_j ; an Edgeworth index h	
σ, ρ, λ	$\sigma_h^2, \sigma_{\text{count}}^2$; $\rho_h, \rho_\ell(d)$; a partition λ	$\sigma = \text{Re } s$; $\rho = t^{3/2}$; standardized cumulants λ_j	
$\varphi, \phi, \varepsilon, \eta$	the characteristic function ϕ_t , the phase φ , $\varphi(k) = \log p(k) - tk$; small constants ε ; $\eta = 24/25$	exponent functions ϕ_0, ϕ_1, ϕ_2 ; $\varepsilon = n^{-1/4}$ and absence probabilities ε_k ; constants η , $\eta_R(Y)$	

3 What neither Part claims

[write] Every limitation stated by either manuscript is printed in its place. Collected:

- *Part I.* An eventual theorem only, not a practical approximation: at $n = 5000$ the leading carrier divided by $a(n)$ is about 0.00177838 and the first-corrected one 0.00486243; a single Gaussian's ratio 1.00047777 there proves nothing. The onset is not effective; there is no finite- n accuracy theorem, no convergence of the expansion and no uniformity in a growing order J ; no reconstruction of every exponentially smaller charge and no convergent transseries. The inverse need not decide the rounding near an integer. The threshold $\alpha = 1/3$ of Section 6 is a modulus observation, not a coefficient theorem. No certified value of C_P . The limit 9/16 is not a finite- M threshold (the equal-weight phase is about 1.02188 at $M = 1000$). The source screen was bounded: no universal novelty claim; the Gingold–Quaintance lead is unverified; the full texts of Garoni–Frankel–Glasser and Leboeuf were not audited. The package has no all-orders symbolic engine and no interval-certified finite- n carrier; its manifest is not a signature and its builder is not a sandbox; its floating diagnostics are uncertified. A preliminary dossier error (a resonance moment without the upper-region restriction) was corrected before delivery (review record, Section 14).

- *Part II.* No effective asymptotic onset; no convergence of the complete formal series (the series H is formal); no bounds uniform in a growing truncation order and no optimal truncation; no interval certificates for asymptotic errors; no unconditional rounding rule and no universal exact floor formula for the inverse; no arbitrary accuracy against the fixed piecewise-linear interpolant (the order- t^4 barrier). The product (Howroyd) and the composition model (Wiseman) are prior; the Jacobi, eta, MacMahon, saddle, Edgeworth and conditioning methods are classical. The source search was bounded and library misses are incomplete; no global priority or “breakthrough” claim; McIntosh, Knopfmacher et al., Kaneiwa and Wright were not read in full. Frozen numerical tables are diagnostics; finite checks prove no asymptotic theorem; the further questions are suggested, not claimed open in the literature; no reproducibility promise across TeX versions.
- *The write.* Its proofs of two outline steps of Part II and its numerical comparisons are unrefereed like the rest. Its statements about other reports are checked against the repository of 5 October 2026.

Part I

Phase-resolved asymptotics for A271619

[write] Part I is Report 182 of the session bundle, *Phase resolved asymptotics for A271619*, author line “A rigorous asymptotic and reproducibility report”, dated 3 October 2026 (30 pages as delivered). The author line is the manuscript’s own description; the manuscript is unrefereed and not formalized. Its abstract, finite-index warning and scope paragraph open this report, before the contents. Its text follows unchanged except for the label prefix **stp:sum:**, one label added to a subsection, the title of Section 15 (was “Further directions”) and the notes marked [write].

4 Model, notation, and main results

An object consists of a finite set $S \subset \mathbb{N}$ of distinct positive outer sizes, together with one ordinary partition of k for each $k \in S$. Its total size is $\sum_{k \in S} k$. Thus

$$a(n) = \sum_{\substack{S \subset \mathbb{N} \text{ finite} \\ \sum S = n}} \prod_{k \in S} p(k), \quad \sum_{n \geq 0} a(n) q^n = \prod_{k \geq 1} (1 + p(k) q^k). \quad (1)$$

The product is analytic for $|q| < 1$ because $p(k) = \exp(O(\sqrt{k}))$. This model allows at most one outer block of each *size*. It is different from $\prod (1 + q^k)^{p(k)}$, which permits distinct types of the same size simultaneously.

Throughout, put

$$b = \pi\sqrt{2/3}, \quad T_m = \frac{m(m+1)}{2}, \quad P_m = \prod_{k=1}^m p(k), \quad M = M(n) = \left\lfloor \frac{\sqrt{8n+1}-1}{2} \right\rfloor. \quad (2)$$

Here n is total size, m is an effective filled-edge charge, and $r_m(x) = x - T_m$ is residual size. Set $c_m(x) = r_m(x)/m$. The symbol J will always denote a fixed expansion order, never a charge. An actual outer cardinality is denoted by K_n . A set of missing low levels is denoted by $\mathcal{H}$; its energy is $h(\mathcal{H}) = \sum_{k \in \mathcal{H}} k$. The tame-edge cutoff is D , and the low-level cutoff is L , avoiding confusion between a cutoff and hole energy.

For sufficiently large real y , define

$$f(y) = b\sqrt{u} - \log(4\sqrt{3}u) + \log\left(1 - \frac{1}{b\sqrt{u}}\right), \quad u = y - \frac{1}{24}. \quad (3)$$

This is the logarithm of the dominant exponential branch of the first Rademacher summand, not the whole hyperbolic-sine summand. We prove below that

$$\log p(k) = f(k) + O(e^{-c\sqrt{k}}), \quad f^{(j)}(y) = O_j(y^{1/2-j}) \quad (j \geq 1). \quad (4)$$

Each derivative has a computable expansion in half-integral powers. The exact low weights remain in P_m .

The limiting hole set $\mathcal{H}_\infty$ has independent memberships

$$\mathbb{P}(k \in \mathcal{H}_\infty) = \frac{1}{p(k) + 1}, \quad C_0 = \prod_{k \geq 1} \left(1 + \frac{1}{p(k)}\right). \quad (5)$$

It is finite almost surely. Write $h_\infty = \sum_{k \in \mathcal{H}_\infty} k$ and $\mu_j = \mathbb{E}h_\infty^j$, with $\mu_0 = 1$. All these moments are finite; no positive ordinary exponential moment exists.

Choose a nonnegative C^∞ function χ , compactly supported in $(1/4, 9/4)$ and equal to one on $[1/2, 7/4]$. Section 8 constructs smooth, computable functions $A_k(c)$ with

$$A_0(c) = 1, \quad A_1(c) = b \left(\frac{21c^2}{80} + \frac{c^{5/2}}{5} \right). \quad (6)$$

For large real x , define the finite sums

$$E_J(x) = \sum_{m \geq 1} \chi(c_m(x)) P_m e^{f'(m+1/2)r_m(x) + f(r_m(x))} \sum_{k=0}^J A_k(c_m(x)) m^{-k/2}, \quad (7)$$

$$\mathcal{A}_J(x) = C_0 \sum_{j=0}^J \frac{\mu_j}{j!} E_{J-j}^{(j)}(x). \quad (8)$$

Only boundedly many m occur at each x ; summands are defined to be zero outside the cutoff support. On that support all arguments of f tend to infinity. Consequently this convention is smooth and creates no small-argument ambiguity.

Theorem 4.1 (All fixed relative orders). *For every fixed integer $J \geq 0$, as $n \rightarrow \infty$ through integers,*

$$a(n) = \mathcal{A}_J(n) \left(1 + O_J(M^{-(J+1)/2})\right). \quad (9)$$

The error is uniform through triangular boundaries and the two-charge crossover. The carrier is eventually positive and

$$0 < c_J M(x)^{-1/2} \leq (\log \mathcal{A}_J)'(x) \leq C_J M(x)^{-1/2} \quad (10)$$

for all sufficiently large real x . The constants and onset may depend on J and χ . No convergence of the infinite expansion and no uniformity in growing J are claimed.

At integer inputs, its all-algebraic-order content comes from charges M and $M - 1$. Write

$$W_m(n) = P_m e^{f'(m+1/2)r_m(n)} p(r_m(n)). \quad (11)$$

When r_M is small, its charge is exponentially subdominant and can be omitted. The first complete correction is

$$B_1(c) = b \left[\frac{21c^2}{80} + \frac{c^{5/2}}{5} + \frac{\mu_1}{2} (1 + c^{-1/2}) \right]. \quad (12)$$

Uniformly, with errors relative to the positive leading sum,

$$a(n) = C_0 \sum_{\substack{m=M, M-1 \\ r_m \geq m/4}} W_m(n) \left[1 + \frac{B_1(c_m)}{\sqrt{m}} + O(M^{-1}) \right]. \quad (13)$$

Dropping the small-residual term costs only an exponentially small relative error. A second explicit coefficient and relative error $O(M^{-3/2})$ are proved in Sections 8.6 and 10.4.

Theorem 4.2 (Discrete inverse). *Let $N(Y) = \min\{n \geq 0 : \log a(n) \geq Y\}$. For fixed J , let $x_J(Y)$ be the unique sufficiently large solution of $\log \mathcal{A}_J(x) = Y$. There is a fixed $D_J > 0$ such that, for all sufficiently large Y ,*

$$\left[x_J(Y) - D_J M(x_J(Y))^{-J/2} \right] \leq N(Y) \leq \left[x_J(Y) + D_J M(x_J(Y))^{-J/2} \right]. \quad (14)$$

At $Y = \log a(k)$, one also has $|x_J(Y) - k| = O_J(M(k)^{-J/2})$.

For $J = 0$ the pre-rounding uncertainty is bounded; for $J \geq 1$ it tends to zero. Near an integer, the theorem need not decide the rounding. It concerns integer-point errors and the explicit carrier slope, not an arbitrary interpolation of the sequence.

Meaning of the phase expansion. The competing exponential charge weights, each carrying its own algebraic series, form a two-sector asymptotic transseries in the fixed-order sense made precise by (9). We do not claim a reconstruction of every exponentially smaller charge or a convergent transseries.

Proof architecture. We first establish elementary growth and the resonance obstruction. The exact filling identity then separates low holes without double counting. The local expansion applies only to tame edge diagrams. A separate global proof controls non-tame hooks and every other charge, paying for each hook's coordinate entropy with its own Bellman gap. Finally, a two-stage weighted hole truncation and a finite Taylor expansion assemble the carrier. This order avoids circular tail estimates and avoids differentiating any unknown coefficient remainder.

5 Elementary growth and classical partition input

Hardy–Ramanujan and Rademacher give [4]

$$\log p(k) = b\sqrt{k} - \log k - \log(4\sqrt{3}) + O(k^{-1/2}). \quad (15)$$

For completeness, the exponentially accurate version (4) follows directly from the convergent Rademacher series. With $u = k - 1/24$, $v = b\sqrt{u}/j$, and the elementary bound $|A_j(k)| \leq j$ on its exponential sum, use

$$v \cosh v - \sinh v = \int_0^v y \sinh y \, dy \leq v^3 e^v / 3.$$

Every $j \geq 2$ summand is $O(j^{-3/2}e^{b\sqrt{u}/j})$; their sum is $O(e^{b\sqrt{u}/2})$. The dominant $j = 1$ branch has order $u^{-1}e^{b\sqrt{u}}$, so the relative error is $O(ue^{-b\sqrt{u}/2})$, hence $O(e^{-c\sqrt{u}})$. The opposite exponential in the first summand is smaller still. Formula (3) and its differentiated expansions follow by elementary differentiation.

Lemma 5.1. $a(n+1) > a(n)$ for $n \geq 1$, while $a(0) = a(1) = 1$.

Proof. Append a part 1 to the inner partition decorating the largest outer size. This raises that size by one, preserves strictness, and is injective: the unique largest block identifies the inverse operation. A single inner partition of $n+1$ having no part 1, for example $(n+1)$ when $n \geq 1$, is outside the image. Thus the inequality is strict. $\square$

Put $F(t) = \log \prod_{k \geq 1} (1 + p(k)e^{-tk})$. Since $|\log(1 + e^u) - u_+| \leq \log 2$ and $|(u+v)_+ - u_+| \leq |v|$, splitting at a sufficiently large constant multiple of b^2/t^2 gives

$$\begin{aligned} F(t) &= \sum_{k \geq 1} (b\sqrt{k} - tk)_+ + O(t^{-2} \log(1/t)) \\ &= \frac{b^4}{6t^3} + O(t^{-2} \log(1/t)). \end{aligned} \tag{16}$$

Indeed the logarithmic errors in (15) sum to the stated bound below that cutoff, the tails decay geometrically above it, and the sum/integral error of the leading unimodal summand is $O(t^{-1})$.

Proposition 5.2.

$$\log a(n) = \frac{2b}{3}(2n)^{3/4} + O(\sqrt{n} \log(n+2)). \tag{17}$$

Consequently, with $C = (2b/3)2^{3/4} = 4\pi^{1/4}/(3\sqrt{3})$,

$$N(Y) = (Y/C)^{4/3} \left(1 + O(Y^{-1/3} \log Y)\right).$$

Proof. Positivity gives $a(n) \leq e^{F(t)+nt}$. Choose $t = b/(2n)^{1/4}$ in (16). For the lower bound, take outer sizes $1, \dots, M$ and add $n - T_M$ to the largest size. Their total weight is at least P_M because p is nondecreasing. Summing (15) gives $\log P_M = (2b/3)M^{3/2} + O(M \log M)$, which matches the upper bound since $M = \sqrt{2n} + O(1)$. Monotonicity and inversion of the leading power give the final formula. $\square$

The elementary exact computation uses descending product updates

$$a^{(k)}(n) = a^{(k-1)}(n) + p(k)a^{(k-1)}(n-k), \quad a^{(0)}(0) = 1.$$

Its independent logarithmic-derivative recurrence is recorded in Section 13.

6 Why a single central minor arc estimate fails

The following concerns characteristic-function moduli, not by itself a coefficient formula. Let $X = X(t)$ be the large solution of $f(X) = tX$, and put $H_t = (t - f'(X))^{-1}$. Then

$$X = \frac{b^2}{t^2}(1 + O(t \log(1/t))), \quad H_t = \frac{2}{t}(1 + O(t \log(1/t))). \tag{18}$$

Use X , not the integer charge M , for this physical crossing. Independent Bernoulli variables B_k with odds $p(k)e^{-tk}$ produce total size $S_t = \sum k B_k$. Write $v_k = \mathbb{P}(B_k = 1)\mathbb{P}(B_k = 0)$ and $\phi_t(\theta) = \mathbb{E}e^{i\theta S_t}$. Exactly,

$$\log |\phi_t(\theta)| = \frac{1}{2} \sum_{k \geq 1} \log \left(1 - 4v_k \sin^2(k\theta/2) \right). \quad (19)$$

In the upper-edge window $k = X + H_t y$, the variance profile tends to $g(y) = e^y/(1 + e^y)^2$, with uniformly exponentially decaying tails after fixed-window enlargement. Taylor expansion of $f(k) - tk$ around X proves this, using $H_t/X \rightarrow 0$. Since $\int_{\mathbb{R}} y^2 g(y) dy = \pi^2/3$,

$$\sum_{k \geq \epsilon X} (k - X)^2 v_k \sim H_t^3 \pi^2/3, \quad \sum_{k \geq \epsilon X} (k - X)^4 v_k = O(H_t^5).$$

The restriction to $k \geq \epsilon X$ is essential: persistent low holes would make the unrestricted centered second moment of order X^2 , larger than the upper-edge term. At $\theta = 2\pi\ell/X$, the sine in (19) equals in modulus $\sin(\pi\ell(k - X)/X)$. Expanding sine and logarithm only in this upper region therefore gives, after adding the lower-region estimate below, for each fixed nonzero integer ℓ ,

$$\log |\phi_t(2\pi\ell/X)| = -\frac{2\pi^4}{3} \ell^2 \frac{H_t^3}{X^2} (1 + o(1)) = -12\ell^2 t (1 + o(1)). \quad (20)$$

Here the lower boundary has not been forgotten. For a sufficiently small fixed $\epsilon > 0$, $v_k \leq C e^{-c\sqrt{k}}$ on $k \leq \epsilon X$, apart from finitely many harmless terms. Thus $\sum_{k < \epsilon X} k^2 v_k = O(1)$. Using the original sine $\sin(\pi\ell k/X)$ there gives modulus contribution $O(\ell^2 X^{-2}) = O(\ell^2 t^4)$, uniformly for $|\ell| \leq C t^{-1/2}$. The centered Taylor approximation is not used in that lower region. The intermediate regions are exponentially small. In the upper edge, fourth moments control the sine/logarithm remainders. Uniformly on $|\ell| \leq C t^{-1/2}$, the relative profile error is $O(t \log(1/t))$ and the quartic error is $O(\ell^4 t^3)$.

Noncentral resonances thus approach modulus one, and on the modulus scale there are order $t^{-1/2}$ relevant resonances. In particular a proposed bound tending uniformly to zero outside a single central saddle is false. This statement alone does *not* logically rule out every possible Gaussian coefficient equivalent; the phase-resolved coefficient theorem requires the independent arguments below. The exceptionally good finite Gaussian fit in the opening warning does not repair the failed minor-arc hypothesis.

A bounded general observation. If a smooth stretched amplitude satisfies $\log w(k) \sim b k^\alpha$, where $b > 0$ and $0 < \alpha < 1$, with sufficiently controlled differentiated remainders and negligible lower-edge contributions on the scale in question, the same calculation gives

$$\log |\phi(2\pi\ell/X)| \sim -\frac{2\pi^4}{3b^3(1-\alpha)^3} \ell^2 X^{1-3\alpha}.$$

Here $t \sim b X^{\alpha-1}$ and $H_t \sim X^{1-\alpha}/(b(1-\alpha))$. This identifies $\alpha = 1/3$ as a natural *modulus* threshold. No general coefficient crossover or all-orders theorem for such weights is asserted.

[write] (5 October 2026, batch 100.) This observation is listed as question (i) of Section 15, with what a proof would need. At $\alpha = 1/2$ it reduces to (20): with $X \sim b^2/t^2$, $X^{-1/2} \sim t/b$ and $b^4 = 4\pi^4/9$, the exponent is $-16\pi^4 \ell^2 t/(3b^4) = -12\ell^2 t$ (checked in the write).

7 The exact low hole filling identity

Fix n , $M = M(n)$, and $L = \lfloor M^{17/20} \rfloor$. Keep this L fixed even when a hole shift changes the total size. For an outer set S define

$$\mathcal{H} = [1, L] \setminus S, \quad h = \sum_{k \in \mathcal{H}} k, \quad S^+ = S \cup [1, L], \quad m = |S^+| = |S| + |\mathcal{H}|. \quad (21)$$

Sort S^+ decreasingly as $k_1 > \dots > k_m$ and set $\lambda_i = k_i - (m - i + 1)$. Then λ is an ordinary partition of $n + h - T_m$ and $\ell(\lambda) \leq m - L$. Conversely every triple

$$\mathcal{H} \subset [1, L], \quad m \geq L, \quad \lambda \vdash n + h - T_m, \quad \ell(\lambda) \leq m - L$$

reconstructs $S^+ = \{m - i + 1 + \lambda_i : 1 \leq i \leq m\}$ and then $S = S^+ \setminus \mathcal{H}$. The length restriction ensures that S^+ contains $[1, L]$, so its missing-low-level set is precisely $\mathcal{H}$. These are mutually inverse maps on outer selections. In particular the filled charge is unique, adjacent charge sectors are disjoint, and the actual cardinality is $m - |\mathcal{H}|$.

Let $A_L(N)$ be the weighted sum (1) restricted to selections containing $[1, L]$, and put $p(\mathcal{H}) = \prod_{k \in \mathcal{H}} p(k)$. The exact weighted identity is

$$a(n) = \sum_{\mathcal{H} \subset [1, L]} \frac{A_L(n + h(\mathcal{H}))}{p(\mathcal{H})}. \quad (22)$$

This is a weighted identity for decorated objects: the new decorations introduced by filling contribute $p(\mathcal{H})$, which is divided out. No literal unweighted bijection to arbitrarily decorated filled objects is needed.

For completeness, define

$$B_{m,r;L} = \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) \leq m-L}} \prod_{i=1}^m \frac{p(m - i + 1 + \lambda_i)}{p(m - i + 1)}, \quad B_{m,r;L} = 0 \quad (r < 0).$$

Then $A_L(N) = \sum_{m \geq L} P_m B_{m,N-T_m;L}$, a finite sum. If (a_i, b_i) are Frobenius arms and legs of λ , its exact weight ratio is

$$\prod_i \frac{p(m + a_i + 1)}{p(m - b_i)}, \quad r = \sum_i (a_i + b_i + 1). \quad (23)$$

Every changed denominator is above L . Thus, after coarse localization $m \asymp M$, replacing changed logarithms by f costs at most $O(me^{-c\sqrt{L}})$ in the log weight. The untouched product P_m remains exact.

7.1 Initial weighted hole truncation without a carrier assumption

Set $t = f(M)/M \sim b/\sqrt{M}$ and $\varphi(k) = \log p(k) - tk$. The exact product identity gives

$$\begin{aligned} F(t) + nt - \log P_M &= t(n - T_M) + \sum_{k \leq M} \log(1 + e^{-\varphi(k)}) \\ &\quad + \sum_{k > M} \log(1 + e^{\varphi(k)}). \end{aligned} \quad (24)$$

The right side is $O(\sqrt{M})$. Here are details covering all ranges. For a small fixed ϵ and sufficiently large $k \leq \epsilon M$, (15) gives $\varphi(k) \geq c\sqrt{k}$; finitely many smaller k contribute $O(1)$. The corrections

are summable. On $[\epsilon M, M/2]$, the leading expression $b\sqrt{M}(\sqrt{k/M} - k/M)$ is at least $c\sqrt{M}$, so its correction is exponentially small. On $[M/2, CM]$, for fixed large C , $f'(x) - t \leq -c/\sqrt{M}$ uniformly. Since $f(M) - tM = 0$, the relevant side of $\varphi(k)$ is at most $-c|k - M|/\sqrt{M}$, up to an exponentially tiny error. Summing gives $O(\sqrt{M})$. Finally, for $k \geq CM$, the bound $\log p(k) \leq b\sqrt{k} \leq tk/2$ gives a geometric tail. Also $t(n - T_M) = O(\sqrt{M})$.

Positivity implies $A_L(n + h) \leq a(n + h) \leq e^{F(t) + t(n+h)}$. For $K = M^\eta$ with $\eta = 24/25 = 0.96$, the contribution of $h > K$ in (22) is therefore at most

$$P_M e^{C\sqrt{M}} \sum_{\substack{\mathcal{H} \subset [1, L] \\ h > K}} \frac{e^{th}}{p(\mathcal{H})}.$$

Take $\lambda = b/(8\sqrt{L})$. Since $t\sqrt{L} \rightarrow 0$, eventually $(t + \lambda)k \leq (b/2)\sqrt{k}$ for every $k \leq L$. The lower bound implicit in (15) makes

$$\prod_{k \leq L} \left(1 + \frac{e^{(t+\lambda)k}}{p(k)} \right) \leq C$$

uniformly: its logarithm is dominated by a convergent sum of $Cke^{-b\sqrt{k}/2}$ plus finitely many factors. Chernoff's inequality bounds the omitted contribution by

$$CP_M \exp(C\sqrt{M} - cM^{\eta-17/40}). \quad (25)$$

Since $\eta - 17/40 = 0.535 > 1/2$, this is smaller than P_M times every fixed negative power of M . This argument uses only the elementary lower bound $a(n) \geq P_M$, not the desired carrier. For $h \leq M^\eta = o(M)$ the continuous triangular root changes by $O(h/M) = o(1)$; its floor remains M or becomes $M + 1$.

8 The tame edge expansion to every fixed order

For a partition λ , set

$$Q_j(\lambda) = \sum_{i \geq 1} [(\lambda_i - i + 1/2)^j - (-i + 1/2)^j], \quad d(\lambda) = \max(\lambda_1, \ell(\lambda)). \quad (26)$$

In particular $Q_1 = |\lambda|$. Fix $1/2 < \delta < 1$ and $D = \lfloor m^\delta \rfloor$. A partition is called tame when $d(\lambda) \leq D$. For $x_m = m + 1/2$, define

$$R_{m,r}^{(D)} = \frac{1}{p(r)} \sum_{\substack{\lambda \vdash r \\ d(\lambda) \leq D}} \exp \left\{ \sum_i [f(m+1-i+\lambda_i) - f(m+1-i)] - f'(x_m)r \right\}.$$

Theorem 8.1 (Local expansion with controlled remainders). *Uniformly for $c = r/m$ in any fixed compact subset of $(0, \infty)$, for every fixed $J \geq 0$,*

$$R_{m,r}^{(D)} = \sum_{k=0}^J A_k(c) m^{-k/2} + O_J(m^{-(J+1)/2}). \quad (27)$$

The coefficients are smooth, effectively computable, independent of δ , and satisfy (6). The same expansion holds for actual partition-number weights after multiplication by $P_m e^{f'(x_m)r} p(r)$.

The theorem is expressly local: it does not apply to an unrestricted cardinality- m sector for all fixed positive c . We exhibit the obstruction after its proof.

8.1 Bernoulli coordinates and the conditioning cost

For positive half-integers h , let U_h, V_h be mutually independent Bernoulli variables of probability $\rho_h = (1 + e^{\tau h})^{-1}$, with $\tau = \pi/\sqrt{6}r$. They have almost surely finite support. Define charge and energy

$$C = \sum_h (U_h - V_h), \quad N = \sum_h h(U_h + V_h).$$

Given $C = 0$, the equal-sized particle and hole sets are exactly Frobenius coordinates of a partition. Given also $N = r$, their common unnormalized weight is $e^{-\tau r}$, so the law is uniform on partitions of r . Exactly on this event,

$$Q_j = \sum_h h^j [U_h + (-1)^{j+1} V_h], \quad Q_j(\lambda') = (-1)^{j+1} Q_j(\lambda). \quad (28)$$

The half-integer energy causes no omitted lattice factor: charge zero forces integral energy.

Write $\mathcal{P}(q) = \prod_{k \geq 1} (1 - q^k)^{-1}$ and $A = \pi^2/6$. Jacobi's triple product gives the exact normalization

$$Z(\tau) = \prod_{h > 0} (1 + e^{-\tau h})^2 = \mathcal{P}(e^{-\tau}) \sum_{a \in \mathbb{Z}} e^{-\tau a^2/2}. \quad (29)$$

Eta and Gaussian transformations give $Z(\tau) \sim e^{A/\tau}$. Hardy–Ramanujan then yields

$$\mathbb{P}(C = 0, N = r) = \frac{p(r)e^{-\tau r}}{Z(\tau)} \sim \frac{1}{4\sqrt{3}r}. \quad (30)$$

For a nonnegative variable Y , therefore $\mathbb{E}_r Y \leq Cr \mathbb{E} Y$. More flexibly, Hölder gives $\mathbb{E}_r |Y| \leq C_q r^{1/q} \|Y\|_q$ for any fixed $q > 1$. We shall choose high fixed moments to reduce this conditioning penalty to any desired small power.

8.2 Exponential moments of the centered tame exponent

For $h \leq D - 1/2$, put

$$\begin{aligned} a(h) &= f(x_m + h) - f(x_m) - f'(x_m)h, \\ d(h) &= -f(x_m - h) + f(x_m) - f'(x_m)h, \\ S_D &= \sum_{0 < h \leq D-1/2} [a(h)U_h + d(h)V_h]. \end{aligned}$$

On the tame conditioning event this is the exact centered exponent. Outside that event it is just a truncated additive random variable, not the true exponent of an unrestricted large partition. Since $D = o(m)$, Taylor's theorem gives uniformly

$$|a(h)| + |d(h)| \leq Cm^{-3/2}h^2, \quad |a(h) + d(h)| \leq Cm^{-5/2}h^3. \quad (31)$$

For every fixed real v , $(|va| + |vd|)/(\tau h) = O_v(h/m) = o(1)$ on this cutoff. Expand the independent Bernoulli logarithmic moment generating functions. Their linear terms pair; their sum is bounded by $C_v m^{-5/2} \sum_h \rho_h h^3 = O_v(m^{-5/2} \tau^{-4}) = O_v(m^{-1/2})$. The quadratic and higher remainder is bounded by $C_v m^{-3} \sum_h h^4 e^{-\tau h/2} = O_v(m^{-3} \tau^{-5}) = O_v(m^{-1/2})$. Thus

$$\mathbb{E} e^{vS_D} = \exp(O_v(m^{-1/2})). \quad (32)$$

In particular $\mathbb{E} e^{v|S_D|}$ is bounded, using $e^{v|s|} \leq e^{vs} + e^{-vs}$.

The first cumulant is $O(m^{-1/2})$. For $s \geq 2$, the Bernoulli bound $|\kappa_s(\text{Bernoulli}(\rho))| \leq C_s \rho$ gives

$$|\kappa_s(S_D)| \leq C_s m^{-3s/2} \sum_h h^{2s} e^{-\tau h} = O_s(m^{-(s-1)/2}).$$

In the moment-cumulant formula for a $2k$ th moment, pairings have the slowest decay, $O(m^{-k/2})$; every other partition of the cumulant indices decays at least as fast. Hence

$$\mathbb{E}|S_D|^{2k} = O_k(m^{-k/2}). \quad (33)$$

All these bounds also hold for any fixed Taylor-polynomial replacement of a, d starting at degree two. Its quadratic bound is unchanged and the even Taylor terms still cancel in $a + d$.

8.3 Finite Taylor reduction with arbitrary accuracy

The independent probability of a particle or hole above $D - 1/2$ is $O(\tau^{-1}e^{-\tau D})$. Combining Cauchy–Schwarz, (32), and (30) shows

$$\mathbb{E}_r[e^{S_D} \mathbf{1}_{d(\lambda) > D}] = O(m^{5/4} e^{-\kappa m^{\delta-1/2}}). \quad (34)$$

This comparison deliberately uses the truncated S_D outside the tame event.

Taylor expansion of e^{S_D} through degree K has conditional remainder at most

$$C_K r (\mathbb{E}|S_D|^{2K+2})^{1/2} (\mathbb{E}e^{2|S_D|})^{1/2} = O_K(m^{1-(K+1)/4}).$$

Replacing f by its Taylor polynomial through degree J_0 introduces an additive error bounded by

$$C_{J_0} m^{-J_0-1/2} \sum_h h^{J_0+1} (U_h + V_h).$$

Every fixed L^s norm of this variable is $O_{J_0,s}(m^{(1-J_0)/2})$. For example Minkowski and $\sum h^{J_0+1} \rho_h^{1/s} = O_s(\tau^{-J_0-2})$ give this directly. The exponential mean-value inequality and Cauchy–Schwarz, followed by conditioning, give expectation error $O_{J_0}(m^{(3-J_0)/2})$.

Finally, a fixed polynomial moment in truncated Frobenius sums differs from the full Q_j moment by a polynomial in m times $e^{-\kappa m^{\delta-1/2}}$, by the same tail event, Hölder, and higher fixed moments. Thus for any requested $O(m^{-B})$ error, it suffices to choose fixed integers

$$J_0 > 2B + 3, \quad K > 4B + 3,$$

and use the finite expression

$$\sum_{k=0}^K \frac{1}{k!} \mathbb{E}_r \left(\sum_{j=2}^{J_0} \frac{f^{(j)}(x_m)}{j!} Q_j \right)^k. \quad (35)$$

No infinite exponential series is interchanged with an uncontrolled partition sum.

8.4 Quasimodular moments and coefficient extraction

For a partition statistic G , write $\langle G \rangle_q = \mathcal{P}(q)^{-1} \sum_{\lambda} G(\lambda) q^{|\lambda|}$. Bloch–Okounkov’s theorem [5] applies to the corrected shifted powers

$$\widehat{Q}_j = Q_j + (1 - 2^{-j}) \zeta(-j), \quad \text{weight}(\widehat{Q}_j) = j + 1. \quad (36)$$

Every fixed polynomial bracket is a finite mixed-weight polynomial in the normalized Eisenstein series E_2, E_4, E_6 . Their explicit n -point determinant formula supplies an effective construction of these polynomials. We do not assume an unspecified number of Fourier coefficients always suffices to determine an arbitrary bracket.

At $q = e^{-z}$ its modular transformation has a finite Laurent growth part $\sum_{d \geq 0} h_d z^{-d}$, with exponentially small corrections in the narrow sector used below. On the whole circle $|q| = e^{-\tau}$, the absolute Fourier/Lambert series bounds any fixed Eisenstein polynomial by a fixed power of τ^{-1} .

This growth expansion is transferred by a genuine coefficient argument. The product satisfies

$$\begin{aligned} \log \mathcal{P}(e^{-\tau}) - \log |\mathcal{P}(e^{-\tau+i\theta})| &\geq \sum_{k \geq 1} e^{-\tau k} (1 - \cos k\theta) \\ &\gg \frac{\theta^2}{\tau(\tau^2 + \theta^2)}, \quad |\theta| \leq \pi. \end{aligned} \quad (37)$$

The first inequality follows by retaining the first term in the logarithmic product series, all terms being nonnegative; the final sum is an explicit difference of geometric sums, with $1 - \cos \theta \asymp \theta^2$ on this interval. Outside $|\theta| \leq \tau^{1+\epsilon}$, where $0 < \epsilon < 1/2$, this gives suppression $e^{-\Omega(\tau^{-1+2\epsilon})}$, absorbing every fixed Eisenstein and normalization power. Within that arc, modular remainders are uniformly exponentially small, and the width exceeds the Gaussian width $\tau^{3/2}$.

Set $R_r = r - 1/24$ and $z_r = 2\sqrt{AR_r}$. The eta expansion $\mathcal{P}(e^{-z}) \sim \sqrt{z/(2\pi)} \exp(A/z - z/24)$ shows that a monomial z^{-d} has local integrand

$$(2\pi)^{-1/2} z^{1/2-d} \exp(A/z + R_r z).$$

Substitution $z = \sqrt{A/R_r} w$ identifies its local steepest-descent expansion with that of the standard Hankel–Schläfli contour for $I_{d-3/2}(z_r)$ [6]. To make this interpretation precise, take the unit circle in w together with both sides of the negative ray from -1 to $-\infty$. On the circle the real exponent is $z_r \cos \theta$; outside a shrinking positive-saddle arc there is superpolynomial attenuation relative to e^{z_r} . On the negative rays the exponent is at most $-z_r$. The short bridges deforming the original central vertical arc to the circle retain a fixed fraction of the endpoint saddle drop. Thus the two *local expansions* agree to all algebraic orders. We do not identify a finite arc exactly with a Bessel function, or use an unregularized possibly nonconvergent vertical integral at $d = 0$.

Dividing by the identical $d = 0$ partition normalization gives the all-algebraic-order rule

$$z^{-d} \mapsto \left(\frac{R_r}{A} \right)^{d/2} \frac{I_{d-3/2}(z_r)}{I_{-3/2}(z_r)}. \quad (38)$$

Fixed-order Bessel expansions with controlled remainders therefore produce full half-power expansions for every $\mathbb{E}_r G$. The orders are half-integral; $I_{-3/2}$ and $I_{3/2}$ differ only beyond algebraic order relative to their growing branch. This is consistent with the prefactor $z^{1/2}$ and with (3).

8.5 Finiteness at each order and the first coefficient

Under the independent law, fixed L^s norms satisfy

$$\|Q_j\|_s = \begin{cases} O_s(r^{j/2+1/4}), & j \text{ even}, \\ O_s(r^{(j+1)/2}), & j \text{ odd}. \end{cases}$$

For even j the sums are centered, so cumulant bounds give the first line; for odd j the positive mean dominates. Higher-moment Hölder makes the conditioning loss at most any prescribed m^ϵ .

For multiplicities α_j define

$$v(\alpha) = \sum_{j \text{ even}} (j/2 - 3/4)\alpha_j + \sum_{\substack{j \geq 3 \\ j \text{ odd}}} (j/2 - 1)\alpha_j.$$

The derivative-weighted monomial has size $O(m^{-v(\alpha)+\epsilon})$. Conjugation (28) makes its expectation vanish unless the number of even-indexed factors is even. For the remaining monomials $v(\alpha)$ is a nonnegative half-integer. Since the exact transferred expansions and derivatives contain only half powers, choosing $\epsilon < 1/2$ rules out any term above the valuation bound. Each factor has positive valuation, at least $1/4$, so only finitely many monomials contribute at each fixed half-order. Derivatives remove the sole logarithm in f , so the A_k contain no hidden $\log m$ factors. Together with (35) this proves the existence and uniform remainder in Theorem 8.1.

The first coefficient can be made fully explicit. With

$$E_2 = 1 - 24 \sum_{n \geq 1} \sigma_1(n)q^n, \quad E_4 = 1 + 240 \sum_{n \geq 1} \sigma_3(n)q^n, \quad E_6 = 1 - 504 \sum_{n \geq 1} \sigma_5(n)q^n,$$

quasimodularity and the weight-four and weight-six bases give

$$\langle Q_3 \rangle_q = \frac{5E_2^2 + 2E_4 - 7}{960}, \quad \langle Q_2^2 \rangle_q = \frac{5E_2^3 - 3E_2E_4 - 2E_6}{6480}. \quad (39)$$

The first basis has dimension two and is determined by the constant and q coefficients of $Q_3 + 7/960$; the second has dimension three and is determined by its first three coefficients. The systems are nonsingular. In particular the q^2 numerator in the second identity is 51840, whereas $\sum_{\lambda \vdash 2} Q_2^2 = 8$, fixing the denominator 6480.

At $q = e^{-\tau}$ the growth parts are

$$E_2 = -4\pi^2\tau^{-2} + 12\tau^{-1}, \quad E_4 = 16\pi^4\tau^{-4}, \quad E_6 = -64\pi^6\tau^{-6}.$$

Consequently

$$\begin{aligned} \langle Q_3 \rangle &\sim \frac{7\pi^4}{60}\tau^{-4} - \frac{\pi^2}{2}\tau^{-3} + \frac{3}{4}\tau^{-2} - \frac{7}{960}, \\ \langle Q_2^2 \rangle &\sim \frac{16\pi^4}{45}\tau^{-5} - \frac{4\pi^2}{3}\tau^{-4} + \frac{4}{3}\tau^{-3}. \end{aligned}$$

The controlled transfer gives

$$\mathbb{E}_r Q_3 = \frac{21}{5}r^2 + O(r^{3/2}), \quad \mathbb{E}_r Q_2^2 = \frac{64\sqrt{6}}{5\pi}r^{5/2} + O(r^2).$$

Since $\mathbb{E}_r Q_2 = 0$, the only order- $m^{-1/2}$ terms are

$$\frac{f'''(x_m)}{6}\mathbb{E}_r Q_3 + \frac{f''(x_m)^2}{8}\mathbb{E}_r Q_2^2.$$

Using $f''(x_m) \sim -b/(4m^{3/2})$ and $f'''(x_m) \sim 3b/(8m^{5/2})$ gives exactly (6), with both signs positive.

For actual weights, every changed argument in a tame diagram lies in $[m - D, m + D]$, and there are at most $2D$ such arguments. Equation (4) therefore changes its log weight by $O(De^{-c\sqrt{m}})$. This proves the last part of Theorem 8.1.

8.6 A second explicit local correction

The same controlled calculation gives

$$A_2(c) = \frac{A_1(c)^2}{2} - \frac{9}{10}c^{3/2} - \frac{53}{20}c^2 - \frac{8}{5}c^{5/2}. \quad (40)$$

We give the exact bracket certificates so that the needed moment products are proved rather than assumed to factor heuristically. Let $U = Q_3 + 7/960$. Quasimodularity places $\langle U^2 \rangle$, $\langle Q_2^2 U \rangle$, and $\langle Q_2^4 \rangle$ in the complete homogeneous weight 8, 10, 12 spans of dimensions 4, 5, 7. Their identities are

$$\langle U^2 \rangle = \frac{73}{179200}E_4^2 + \frac{1}{4032}E_2E_6 - \frac{1}{5120}E_2^2E_4 - \frac{5}{12288}E_2^4, \quad (41)$$

$$\begin{aligned} \langle Q_2^2 U \rangle = & -\frac{301}{1555200}E_4E_6 + \frac{49}{1036800}E_2E_4^2 + \frac{119}{622080}E_2^2E_6 \\ & + \frac{119}{1244160}E_2^3E_4 - \frac{35}{248832}E_2^5, \end{aligned} \quad (42)$$

$$\begin{aligned} \langle Q_2^4 \rangle = & \frac{151}{3499200}E_6^2 + \frac{7}{93312}E_4^3 - \frac{149}{1166400}E_2E_4E_6 - \frac{199}{1555200}E_2^2E_4^2 \\ & + \frac{29}{699840}E_2^3E_6 + \frac{37}{233280}E_2^4E_4 - \frac{35}{559872}E_2^6. \end{aligned} \quad (43)$$

These are exact identities in the full known spans, not fits to an unrestricted series. In the displayed basis orders, the matrices of coefficients $q^0, \dots, q^{d-1}$ have determinants

$$\begin{aligned} & -17336861982720, \quad 3559021983729451008000, \\ & -124747187309624027471021601904499097600000. \end{aligned}$$

They are nonzero. Direct enumeration of the first d partition sizes therefore determines the true bracket uniquely in its complete span and gives the displayed coefficients. Independent additional checks through q^{20} are redundant diagnostics. The included exact-rational checker exposes the finite coefficient systems and their determinants.

Modular substitution gives the leading terms

$$\frac{49\pi^8}{3600\tau^8}, \quad \frac{28\pi^8}{675\tau^9}, \quad \frac{256\pi^8}{675\tau^{10}},$$

respectively. The controlled transfer (38) proves

$$\begin{aligned} \mathbb{E}_r Q_3^2 &= \frac{441}{25}r^4 + O(r^{7/2}), \\ \mathbb{E}_r(Q_2^2 Q_3) &= \frac{1344\sqrt{6}}{25\pi}r^{9/2} + O(r^4), \\ \mathbb{E}_r Q_2^4 &= \frac{73728}{25\pi^2}r^5 + O(r^{9/2}). \end{aligned}$$

Removing the anomaly in U affects only smaller orders. Writing $a = 21/5$ and $v = 64\sqrt{6}/(5\pi)$, these constants equal $a^2, av, 3v^2$. Their values have been obtained from exact brackets without an unproved conditional central limit or independence assertion.

The first subleading term of the Bessel rule is

$$\left(\frac{R_r}{A}\right)^{d/2} \frac{I_{d-3/2}(2\sqrt{AR_r})}{I_{-3/2}(2\sqrt{AR_r})} = \left(\frac{r}{A}\right)^{d/2} \left(1 - \frac{d(d-3)}{4\sqrt{Ar}} + O(r^{-1})\right).$$

Combining it with the next Laurent terms in (39) gives

$$\begin{aligned}\mathbb{E}_r Q_3 &= \frac{21}{5}r^2 - \frac{36\sqrt{6}}{5\pi}r^{3/2} + O(r), \\ \mathbb{E}_r Q_2^2 &= \frac{64\sqrt{6}}{5\pi}r^{5/2} - \frac{240}{\pi^2}r^2 + O(r^{3/2}).\end{aligned}$$

Direct differentiation at $x_m = m + 1/2$ gives

$$f''(x_m) = -\frac{b}{4}m^{-3/2} + m^{-2} + O(m^{-5/2}), \quad f'''(x_m) = \frac{3b}{8}m^{-5/2} - 2m^{-3} + O(m^{-7/2}).$$

The valuation/parity argument proves that through order m^{-1} the only surviving terms are

$$\begin{aligned}\frac{f'''}{6}\mathbb{E}Q_3, \quad \frac{(f'')^2}{8}\mathbb{E}Q_2^2, \quad \frac{(f''')^2}{72}\mathbb{E}Q_3^2, \\ \frac{(f'')^2 f'''}{48}\mathbb{E}(Q_2^2 Q_3), \quad \frac{(f'')^4}{384}\mathbb{E}Q_2^4.\end{aligned}$$

The subleading contributions of the first two are respectively $-9c^{3/2}/10 - 7c^2/5$ and $-5c^2/4 - 8c^{5/2}/5$. The last three sum to $A_1(c)^2/2$. This proves (40), with $O(m^{-3/2})$ remainder supplied by the already proved finite Taylor and transfer theorem. Any further nonzero monomial has valuation at least $3/2$.

Why the restriction matters. Adding a full column of height m to a tame partition of $r - m$ gives a family of actual cardinality- m objects with exponential ratio to the putative unrestricted typical term

$$\exp \left\{ b\sqrt{m} \left[\frac{1}{2} + \sqrt{c-1} - \sqrt{c} \right] + o(\sqrt{m}) \right\}.$$

This dominates when $c > 25/16$. Such a column encodes a low hole and a changed edge, and is not bounded by the truncated-exponent tail (34). The global theorem must separate it correctly.

9 Global filled block localization

For total size N , let $M_N = \lfloor (\sqrt{8N+1} - 1)/2 \rfloor$ and $r_m = N - T_m$. Put $L = D = \lfloor M_N^{17/20} \rfloor$. Let $E_m^{\text{tame}}(N; D)$ be the exact-weight cardinality- m sum whose edge partition satisfies $d(\lambda) \leq D$, and define

$$B(N) = \sum_{m=M_N, M_N-1} P_m e^{f'(m+1/2)r_m} p(r_m). \quad (44)$$

This $B(N)$ is a positive comparison normalization, not the first correction $B_1(c)$.

Theorem 9.1 (Filled block localization). *There exist positive constants C, κ such that, for all sufficiently large N ,*

$$A_L(N) = E_{M_N}^{\text{tame}}(N; D) + E_{M_N-1}^{\text{tame}}(N; D) + O(B(N)e^{-\kappa M_N^{7/20}}). \quad (45)$$

The discarded difference is nonnegative. The result is uniform when L, D are instead taken from a reference M with $M_N/M \rightarrow 1$, provided both are comparable to $M_N^{17/20}$ with fixed positive comparison constants.

For the two retained charges and large N , $m - D \geq L$, so their tame diagrams really do contain $[1, L]$. Thus the right side retains genuine disjoint subfamilies. Integer shifts of cutoff conventions only affect constants.

9.1 The Bellman envelope and its equality cases

Define

$$S(c) = c/2 + \sqrt{c}, \quad V(c) = \max_{0 \leq j \leq [c]} \{j + S(c - j)\}, \quad (46)$$

$$G(d) = \begin{cases} 1 - \sqrt{1 - d}, & 0 \leq d \leq 1, \\ \sqrt{d}, & d \geq 1. \end{cases} \quad (47)$$

A hook with hole at ym and particle at $(y + d)m$ has normalized square-root gain $g(y, d) = \sqrt{y + d} - \sqrt{y}$. This decreases in y , and $0 \leq y \leq 1$, $y + d \geq 1$; hence its maximum is $G(d)$.

Write $w(c) = \sqrt{c} - c/2$. It increases up to 1 and decreases thereafter. For $c \geq 1$ only the lattice residuals $u = \{c\}$ and $u + 1$ can maximize, giving

$$V(c) = c + W(\{c\}), \quad W(u) = \max\{w(u), w(u + 1)\}. \quad (48)$$

Their crossing is $u = 9/16$. The periodic extension of W is continuous and Lipschitz, with $15/32 \leq W \leq 1/2$. The dominant residual belongs to $[9/16, 25/16]$ once $c \geq 9/16$. Therefore

$$V = S \text{ on } [0, 25/16], \quad V(c + 1) = V(c) + 1 \quad (c \geq 9/16), \quad (49)$$

$$V(y) - V(x) \geq \frac{9}{10}(y - x) \quad (y \geq x \geq 0). \quad (50)$$

Indeed the a.e. derivative lies in $[9/10, 7/6]$ away from the initial segment, and on the initial segment it is larger than $9/10$. The unbounded derivative at zero is integrable and poses no problem for absolute continuity. Put $Q(c) = V(c) - S(c)$. Then $Q \geq 0$, $Q(c) > 0$ for $c > 25/16$, and $Q(c) \geq c/4$ for $c \geq 16$.

The Bellman inequality is

$$V(c) \geq G(d) + V(c - d), \quad 0 \leq d \leq c. \quad (51)$$

For $0 < d < 1$, choose a maximizing integer j for $V(c - d)$ and let $u = c - j$. The variable part is $G(d) + S(u - d)$. If $u \leq 1$, its derivative is at most $-1/2$; if $u > 1$, its second derivative is $[4(1 - d)^{3/2}]^{-1} - [4(u - d)^{3/2}]^{-1} > 0$. Its maximum therefore lies at an allowed integer endpoint, strictly so for interior d . For $1 < d < 2$, splitting increases the gain because $\sqrt{d} < 2 - \sqrt{2 - d}$; for larger d use $\sqrt{d} < 1 + \sqrt{d - 1}$ repeatedly. The endpoints are controlled by $V(x + 1) \geq 1 + V(x)$. This proves (51); the only possible zero-gap positive hook has $d = 1$.

9.2 A quantitative gap after filling low holes

Let $\ell = L/m$. Because filled selections have no hole below $L + 1$, the constrained envelope is

$$G_\ell(d) = \begin{cases} G(d), & d \leq 1 - \ell, \\ \sqrt{d + \ell} - \sqrt{\ell}, & d \geq 1 - \ell. \end{cases}$$

Lemma 9.2. *For sufficiently small $\ell > 0$, uniformly in $c \geq d \geq 0$,*

$$V(c) - G_\ell(d) - V(c - d) \geq \kappa \rho_\ell(d), \quad \rho_\ell(d) = \min\{d, |d - 1| + \sqrt{\ell}\}. \quad (52)$$

Moreover $\rho_\ell(d) \geq (\sqrt{\ell}/2)d$, and if D/m is comparable to ℓ , every $d \geq D/m$ pays $\rho_\ell(d) \geq \kappa D/m$.

Proof. Near $d = 0$, use (50) and $G(d) = d/2 + O(d^2)$. For $d \geq 2$ the gap is at least $0.9d - \sqrt{d} \geq (0.9 - 1/\sqrt{2})d$. Away from zero and one in $0 < d < 2$, strictness of (51) gives a positive compact minimum. This is uniform in unbounded c : if $c - d \geq 9/16$, subtract $\lfloor c - d - 9/16 \rfloor$ from both arguments of V , preserving the gap by (49) and placing them in a fixed compact range. Otherwise they already lie in a compact range.

The simultaneous limits near $d = 1$ require separate estimates. For $d = 1 - \epsilon$ and $c \geq 1.1$, use $V(c) - V(c - 1) \geq 1$ and uniform Lipschitz bounds on $[0.1, \infty)$. If $\epsilon \geq \ell$, the gap is at least $\sqrt{\epsilon} - C\epsilon$, which dominates $\epsilon + \sqrt{\ell}$ for small ϵ . If $\epsilon \leq \ell$, then $G_\ell(d) \leq 1 + \ell/2 - \sqrt{\ell}$ and the gap is at least $\sqrt{\ell} - (C + 1/2)\ell$. For $c < 1.1$ near $d = 1$, the unfilled gap is bounded below by a positive constant: at $c = d = 1$ it is $1/2$, and on $1 \leq c \leq 1.1$, $V(c) - V(c - 1) - 1$ has a positive minimum; continuity also covers nearby $c < 1$.

For $d = 1 + \epsilon$, the unfilled gap is at least 0.4ϵ , by (50) and $\sqrt{1 + \epsilon} \leq 1 + \epsilon/2$. Filling adds $\sqrt{\ell} - [\sqrt{d + \ell} - \sqrt{d}] \geq \sqrt{\ell} - \ell/(2\sqrt{d}) \geq \sqrt{\ell}/2$. These estimates prove (52). For $d \leq 2$ both terms in the minimum defining ρ_ℓ are at least $(\sqrt{\ell}/2)d$; for $d \geq 2$ use $d - 1 \geq d/2$. Finally, when $D/m \asymp \ell \rightarrow 0$, eventually $\sqrt{\ell} \geq \kappa D/m$; the displayed minimum then proves the last assertion. $\square$

9.3 Exact extraction and a uniform tame core bound

In (23), remove every diagonal hook whose arm or leg violates the tame cutoff. Removing any subset of Frobenius coordinates leaves strictly decreasing arm and leg lists of equal length, hence another partition. Put

$$d_i = (a_i + b_i + 1)/m, \quad D_* = \sum_{\text{removed}} d_i, \quad c = r_m/m, \quad v = c - D_*.$$

Each removed hook has $d_i \geq D/m$ up to one harmless integer endpoint. Its hole exceeds L . Successive applications of Lemma 9.2, at the *same* anchor m , telescope:

$$\sum_{\text{removed}} g(y_i, d_i) + V(v) \leq V(c) - \kappa \sum_{\text{removed}} \rho_\ell(d_i). \quad (53)$$

No anchor-change error occurs. Uniformly for $u > v \geq L$, (15) gives

$$\log \frac{p(u)}{p(v)} = b(\sqrt{u} - \sqrt{v}) - \log(u/v) + O(L^{-1/2}) \leq b(\sqrt{u} - \sqrt{v}) + O(L^{-1/2}).$$

The error is only $O(L^{-1/2})$ per hook; its negative logarithmic prefactor is not an uncontrolled loss.

Let $Z_m(s; D)$ be the exact tame sum of residual energy s , divided by P_m , and put $v = s/m$. We need an upper bound uniformly down to $s = 0$ and for growing v , not just the compact-positive local theorem.

Fix an arbitrary constant $C_* \geq 16$. For $1 \leq s \leq C_*m$, use the Bernoulli representation with $\tau = \pi/\sqrt{6s}$. Equation (30), with finitely many small s absorbed, gives conditioning cost at most $C(s + 1)$. The estimates (31) still give

$$\log \mathbb{E} e^{S_D} \leq C[m^{-5/2}\tau^{-4} + m^{-3}\tau^{-5}] \leq C_{C_*} m^{-1/2}.$$

Indeed $(|a| + |d|)/(\tau h) = O(m^{-3/2}D\sqrt{s}) = O(D/m) = o(1)$ uniformly in this larger endpoint range. Since $f'(m + 1/2) \leq b/(2\sqrt{m})$ eventually, (4) gives

$$Z_m(s; D) \leq C(s + 1)p(s)e^{bs/(2\sqrt{m})} \leq m^{C_1} e^{b\sqrt{m}S(v)}. \quad (54)$$

Here $C_1 > 0$ is a fixed polynomial-loss exponent, and $p(s) \leq e^{b\sqrt{s}}$, proved in the next subsection. At $s = 0$, $Z_m(0; D) = 1$ exactly, closing the endpoint without a degenerate Bernoulli parameter.

For arbitrary tame energy, every hook lies in $[m - D, m + D]$. The square-root mean-value theorem gives

$$\sum_i (\sqrt{u_i} - \sqrt{v_i}) \leq \frac{s}{2\sqrt{m}} + C \frac{D}{m} \frac{s}{\sqrt{m}}.$$

The logarithmic partition corrections are nonpositive apart from $O(m^{-1/2})$ per hook. There are at most $\sqrt{s}$ diagonal hooks. Multiplying by the number $p(s)$ of possible partitions gives

$$\log Z_m(s; D) \leq b\sqrt{m}S(v) + C(D/m)v\sqrt{m} + O(\sqrt{v}). \quad (55)$$

For $v \geq 16$, $Q(v) \geq v/4$, so half of $b\sqrt{m}Q(v)$ absorbs the positive errors uniformly, since $D/m \rightarrow 0$. For $v \leq 16$, use (54). Hence throughout the required range,

$$Z_m(s; D) \leq m^{C_1} \exp\{b\sqrt{m}[V(v) - Q(v)/2]\}. \quad (56)$$

Combining this with exact extraction and (53), the sum for each fixed removed-coordinate list is at most

$$P_m m^{C_1} \exp\left\{b\sqrt{m}\left[V(c) - \kappa \sum_i \rho_\ell(d_i) - Q(v)/2\right]\right\}. \quad (57)$$

The per-hook partition error is absorbed by reducing κ . Ignoring compatibility constraints between the removed list and the core only enlarges the sum; each original selection has one canonical list.

9.4 Coarse cardinality localization with an explicit margin

The elementary bound $p(k) \leq e^{b\sqrt{k}}$ follows from

$$\log \mathcal{P}(e^{-t}) \leq \int_0^\infty -\log(1 - e^{-tx}) dx = A/t$$

and optimization of $p(k) \leq e^{A/t+tk}$. The integral bounds the sum because its summand is positive and decreasing.

For m distinct positive integers of sum N , write them increasingly as $i + a_i$, where a_i is nonnegative and nondecreasing, with mean $a = (N - T_m)/m$. Concavity and opposite-order covariance give

$$\sum_i \sqrt{i + a_i} \leq \sum_i \sqrt{i + a} + \sum_i \frac{a_i - a}{2\sqrt{i + a}} \leq \sum_i \sqrt{i + a}. \quad (58)$$

The second sum is nonpositive because its first sequence increases with zero sum and its second decreases.

Temporarily write $R = M_N$, $j = R - m$, and $r = N - T_R \leq R$. If $m \geq 3R/4$, then

$$a = j + u, \quad u = \frac{r + j(j+1)/2}{m}, \quad \sum_{i=1}^m \sqrt{i + a} \leq \frac{2}{3}[(R + u + 1)^{3/2} - a^{3/2}].$$

Here $a \geq j$ and $u/R \leq 1/24 + o(1)$. The upper-end increment is bounded by

$$\sqrt{R + u + 1}(u + 1) \leq \left(\frac{\sqrt{25/24}}{3} + o(1)\right) j^{3/2} + O(\sqrt{R}),$$

because $\sqrt{j/R}/[2(1-j/R)] \leq 1/3$ for $j/R \leq 1/4$. Its coefficient is less than $0.341 < 2/3$. After weakening the margin,

$$\sum_i \sqrt{i+a_i} \leq \frac{2}{3}R^{3/2} - \frac{1}{4}j^{3/2} + O(\sqrt{R}) \quad (59)$$

uniformly in that range. For $m \leq 3R/4$, Jensen directly gives $\sum \sqrt{k_i} \leq \sqrt{mN} \leq (\sqrt{3/8} + o(1))R^{3/2}$, a fixed gap below $(2/3)R^{3/2}$.

There are at most $p(N) = e^{O(R)}$ selections, whereas $\log P_R = (2b/3)R^{3/2} - O(R \log R)$. Thus all $j \geq R^{7/10}$ contribute at most $P_R e^{-\kappa R^{21/20}}$, after absorbing $O(R \log R)$. This step uses neither filled holes nor a global equivalent. Since $B(N) \geq P_R$ eventually, we may restrict to

$$0 \leq j = R - m \leq R^{7/10}, \quad m \sim R, \quad c = r_m/m = O(R^{7/10}). \quad (60)$$

9.5 Rebasing the envelope to the two-charge carrier

In (60), let $u_0 = r/R$ and $c = j + u$, with u as above. Uniformly including $r = 0$, $\log p(r) = b\sqrt{r} + O(\log(r+2))$. The two carrier logarithms, after subtracting $\log P_R$, are

$$b\sqrt{R}S(u_0) + O(\log R), \quad b\sqrt{R}[S(u_0 + 1) - 1] + O(\log R).$$

Taking the log of their positive sum changes the maximum by at most $\log 2$, so

$$\log B(N) = \log P_R + b\sqrt{R}[u_0 + W(u_0)] + O(\log R). \quad (61)$$

At $r = 0$, $p(0) = 1$ is exact and the other charge supplies the positive $b\sqrt{R}/2$ term; no positive-residual asymptotic is applied to $p(0)$.

Summing (15) over $m+1, \dots, R$ gives

$$\begin{aligned} \log P_R - \log P_m &= bj\sqrt{m} + \frac{bj(j+1)}{4\sqrt{m}} - j \log m \\ &\quad + O(j + j^2/m + j^3/m^{3/2}). \end{aligned}$$

For $j \geq 1$, $V(c) = c + W(u)$ with periodic W , and $|u - u_0| = O((j^2 + j)/R)$. Its Lipschitz bound, together with the square-root changes, proves

$$\log P_m + b\sqrt{m}V(c) \leq \log B(N) + C[(j+1) \log R + j^2/\sqrt{R}]. \quad (62)$$

For $j = 0$ the actual first carrier summand gives the result immediately. In particular the potentially large $j\sqrt{R}$ terms cancel explicitly; they are not hidden in the error. Exact P_m is still retained in the final algebraic carrier.

9.6 Paying for every hook and concluding localization

Within the coarse window every hook coordinate is at most $N = O(R^2)$. A list of q removed hooks therefore has at most R^{Cq} choices. Its total energy fixes the core energy. Reserve a fixed fraction of the loss in (57) for entropy. By Lemma 9.2, every removed hook pays order

$$\sqrt{m}(D/m) = D/\sqrt{m} \asymp R^{7/20},$$

which absorbs its own R^C coordinate choices. Summing over q is then bounded by a geometric majorant. Comparing one isolated hook loss against the entropy of all hook lists would not be a valid substitute.

The remaining loss is coercive in total residual. For sufficiently large fixed c , if $D_* \geq c/2$, use $\sum \rho_\ell(d_i) \geq \kappa\sqrt{\ell}D_*$; otherwise $v \geq c/2$ and $Q(v) \geq v/4$. Thus

$$\sum_i \rho_\ell(d_i) + Q(v) \geq \kappa c\sqrt{\ell}.$$

After multiplication by $\sqrt{m}$, the loss is at least $\kappa c\sqrt{L}$. It dominates the rebasing error uniformly in (60), since

$$\log R = o(\sqrt{L}), \quad \frac{j^2/\sqrt{R}}{j\sqrt{L}} \leq R^{1/5-17/40} = R^{-9/40} \rightarrow 0.$$

Polynomial prefactors and summation over at most R charges are absorbed. Hence all sufficiently large deficits j are negligible.

Only bounded j remain. If any hook was extracted, its surviving per-hook penalty gives $e^{-\kappa R^{7/20}}$, absorbing all polynomial losses. If none was extracted and $j \geq 2$, then $c \geq 2$, and $Q(c)$ has a fixed positive minimum, giving $e^{-\kappa\sqrt{R}}$. With no extracted hook and $j = 0, 1$, the selections are exactly the two retained tame sectors. Their filled constraint was checked above. The coarse tail is smaller still. This proves Theorem 9.1, with nonnegative discarded weights and the asserted uniformity in nearby-reference cutoffs. No effective κ or onset is claimed.

10 Finite moment assembly and smooth carrier bounds

We now prove Theorem 4.1, keeping the order of limits and truncations explicit. Fix the order J , the reference $M = M(n)$, and $L = \lfloor M^{17/20} \rfloor$. The initial truncation in Section 7.1 removes $h > M^{24/25}$ from (22) with a superpolynomial error relative to P_M .

For every retained $N = n + h$, Theorem 9.1 applies uniformly with this same L . The tame theorem supplies the integer-point approximation

$$A_L(N) = E_J(N) + O_J(E_0(N)M^{-(J+1)/2}). \quad (63)$$

Here is why the local theorem's compact-positive scope suffices. The leading phase is $w(c) = \sqrt{c} - c/2$, whose two competing residuals have dominant value in $[9/16, 25/16]$, up to $o(1)$ shifts. The plateau of χ strictly contains this range. Its omitted or transition-support charges have a fixed positive $\sqrt{M}$ -scale gap. The small-residual $m = M_N$ term is exponentially subdominant and its upper bound follows from (54); no local expansion at $r = o(m)$ is required. Thus $E_0(N)$ is uniformly comparable to $B(N)$ even through triangular boundaries. In particular $A_L(n) \geq cB(n)$, and (22) gives $a(n) \geq cB(n)$.

10.1 Direct derivative bounds on explicit functions

For a bulk summand of E_0 , its logarithmic slope is

$$f'(m + 1/2) + f'(r_m) = \frac{b}{2\sqrt{m}}(1 + c_m^{-1/2}) + O(m^{-1}). \quad (64)$$

This is positive and comparable to $M^{-1/2}$ on the dominant compact range. Cutoff derivatives cost $O(M^{-1})$ times exponentially smaller weights. Consequently

$$cM^{-1/2} \leq E'_0(x)/E_0(x) \leq CM^{-1/2}. \quad (65)$$

For each fixed local index k and derivative order d , differentiating its explicit summand gives a bound of size

$$C_{k,d} (\text{corresponding leading weight}) M^{-(k+d)/2}. \quad (66)$$

Here the corresponding leading weight is the uncut factor $P_m \exp\{f'(m+1/2)r_m + f(r_m)\}$, without χ ; no bound of a derivative of χ by χ is assumed. Indeed its first exponential derivative is $O(M^{-1/2})$; the d th higher derivative of its exponent for $d \geq 2$ is $O(M^{1/2-d})$. Product/Bell differentiation therefore gives the asserted scale. Derivatives of $A_k(c_m)$ and $\chi(c_m)$ each cost M^{-1} , a smaller scale. Cutoff terms are absorbed using their phase gap. Summing boundedly many charges yields

$$|E_K^{(d)}(x)| \leq C_{K,d} E_0(x) M^{-d/2} \quad (67)$$

for fixed K, d . These are derivatives of explicitly defined smooth functions; no coefficient remainder is being differentiated.

Integrating (65) and using $E_0 \asymp B$ gives the carrier ratio bound

$$B(n+h) \leq CB(n) \exp(Ch/\sqrt{M}), \quad 0 \leq h \leq M^{24/25}. \quad (68)$$

This now permits summation of the local errors and a much sharper hole truncation.

10.2 The second hole truncation and the absence of an ordinary MGF

For every $v > 0$, the unrestricted law in (5) has

$$\mathbb{E} e^{vh_\infty} = C_0^{-1} \prod_{k \geq 1} (1 + e^{vk}/p(k)) = \infty,$$

since $\log p(k) \sim b\sqrt{k}$. Its moment series is not an analytic positive-argument MGF. The argument below instead uses a finite tilted product and a stretched-exponential moment.

For fixed C and $k \leq L = o(M)$, eventually $Ck/\sqrt{M} \leq (b/4)\sqrt{k}$. Since $\sqrt{\sum k} \leq \sum \sqrt{k}$, for any fixed $0 < c < b/2$ we have

$$\sum_{\mathcal{H} \subset [1,L]} \frac{\exp(Ch/\sqrt{M} + c\sqrt{h})}{p(\mathcal{H})} \leq \prod_{k \leq L} \left(1 + \frac{e^{(b/4+c)\sqrt{k}}}{p(k)}\right) \leq C'. \quad (69)$$

The final bound follows from (15) and summability. It implies all fixed tilted polynomial moment bounds and

$$\sum_{\substack{\mathcal{H} \subset [1,L] \\ h > K}} \frac{e^{Ch/\sqrt{M}}}{p(\mathcal{H})} \leq C' e^{-c\sqrt{K}}. \quad (70)$$

After (68), we may take any small fixed power $K = M^\epsilon$, for example $\epsilon = 1/10$. This removes its complement superpolynomially relative to $B(n)$. The unweighted version also proves finiteness of every μ_j . Extending a fixed polynomial moment from $[1, L]$ to all levels changes it by a polynomial in L times $e^{-c\sqrt{L}}$: split the independent low and high parts and bound a nonempty high part by sums of $k^j/p(k)$.

An equivalent elementary proof of (70) splits sets containing some $k > K$ from those entirely in $[1, K]$. The former are bounded by $C \sum_{K < k \leq L} e^{Ck/\sqrt{M}}/p(k)$. For the latter use the *new* Chernoff parameter $b/(8\sqrt{K})$, not the first-stage parameter involving L . Its finite tilted product is bounded, and it pays $e^{-c\sqrt{K}}$ for energy $h > K$. This makes explicit why the initial and later truncations are distinct arguments.

10.3 Assembly at the correct fixed order

First convolve the pointwise estimate (63) in (22); bounds (68) and (69) control the total error. Then restrict to $h \leq M^\epsilon$ by (70). Taylor-expand the explicit $E_J(n+h)$ through degree J . Equations (67) and (68) bound its remainder by

$$C_J E_0(n) M^{-(J+1)/2} h^{J+1} e^{Ch/\sqrt{M}}.$$

Its sum is bounded by the fixed tilted moment in (69). Extending each of the finitely many moments to the full hole law therefore gives

$$C_0 \sum_{j=0}^J \frac{\mu_j}{j!} E_J^{(j)}(n) + O_J(E_0(n) M^{-(J+1)/2}).$$

In the j th term replace $E_J^{(j)}$ by $E_{J-j}^{(j)}$. Every omitted explicit local index has $k+j \geq J+1$, so (66) gives

$$|(E_J - E_{J-j})^{(j)}| \leq C_J E_0 M^{-(J+1)/2}. \quad (71)$$

This produces exactly (8) and (9). Every sum used here is finite or a fixed controlled moment. There is no interchange with an infinite Taylor series or a nonexistent positive MGF.

Direct termwise estimates further give

$$\mathcal{A}_J = C_0 E_0 (1 + O(M^{-1/2})), \quad \mathcal{A}'_J = C_0 E'_0 (1 + O(M^{-1/2})).$$

Terms with $j \geq 1$ have sizes $O(E_0 M^{-j/2})$ and derivatives $O(E_0 M^{-(j+1)/2})$. Together with (65), this proves eventual positivity and (10). It also shows why a floor-defined two-charge expression can be replaced by this genuinely smooth finite-charge carrier: any extra cutoff charges and their derivatives are exponentially small.

At first order, the local correction is $A_1(c)/\sqrt{m}$. The first hole moment contributes $\mu_1 E'_0$, whose leading relative value is $b\mu_1(1+c^{-1/2})/(2\sqrt{m})$ by (64). This proves (12) and (13). On its retained compact range, replacing $e^{f(r_m)}$ by exact $p(r_m)$ is exponentially accurate.

10.4 The second complete relative correction

Let $\sigma_h^2 = \text{Var}(h_\infty) = \sum_{k \geq 1} k^2 p(k)/(p(k)+1)^2$; this is the variance of hole *energy*, not of hole count. The explicit second coefficient is

$$\begin{aligned} B_2(c) = & \frac{B_1(c)^2}{2} - \frac{9}{10} c^{3/2} - \frac{53}{20} c^2 - \frac{8}{5} c^{5/2} - \mu_1(1+c^{-1}) \\ & + \frac{b^2 \sigma_h^2}{8} (1+c^{-1/2})^2. \end{aligned} \quad (72)$$

In particular (13) strengthens to

$$a(n) = C_0 \sum_{\substack{m=M, M-1 \\ r_m \geq m/4}} W_m(n) \left[1 + \frac{B_1(c_m)}{\sqrt{m}} + \frac{B_2(c_m)}{m} + O(M^{-3/2}) \right], \quad (73)$$

with uniform error relative to the positive leading sum.

To check the formula, differentiate only the smooth replacement $\widetilde{W}_m(y) = P_m \exp\{f'(m+1/2)(y-T_m) + f(y-T_m)\}$. Put

$$d_0(c) = \frac{b}{2}(1+c^{-1/2}), \quad d_1(c) = -(1+c^{-1}).$$

Uniformly in the retained compact range,

$$\begin{aligned}(\log \widetilde{W}_m)' &= d_0 m^{-1/2} + d_1 m^{-1} + O(m^{-3/2}), \\ \widetilde{W}_m''/\widetilde{W}_m &= d_0^2 m^{-1} + O(m^{-3/2}), \\ [\widetilde{W}_m(1 + A_1(c)m^{-1/2})]'/\widetilde{W}_m &= d_0 m^{-1/2} + (d_1 + d_0 A_1)m^{-1} + O(m^{-3/2}).\end{aligned}$$

The derivative of $A_1(c)m^{-1/2}$ is $A_1'(c)m^{-3/2}$ and does not enter the displayed second coefficient. The order-two finite carrier is $C_0(E_2 + \mu_1 E_1' + \mu_2 E_0''/2)$, where these E_j denote the charge carriers (7), not Eisenstein series. It follows that

$$B_2 = A_2 + \mu_1(d_1 + d_0 A_1) + \frac{\mu_2}{2} d_0^2.$$

Using $B_1 = A_1 + \mu_1 d_0$ and $\mu_2 = \mu_1^2 + \sigma_h^2$ yields (72). All interchanges and remainders are covered by the finite assembly above. This extra term does not assert good finite-index accuracy or an effective onset.

11 Crossover, product normalization, and inversion

11.1 The slowly moving transition

For a physical cell $T_M \leq n < T_{M+1}$, put $\varphi = (n - T_M)/M$. At integer inputs, $0 \leq \varphi \leq 1$. Near $\varphi_0 = 9/16$, the two leading weights are

$$W_0 = P_M e^{f'(M+1/2)\varphi M} p(\varphi M), \quad W_1 = P_{M-1} e^{f'(M-1/2)(\varphi+1)M} p((\varphi+1)M).$$

For real phase calculations use e^f in place of p . Expanding their log ratio gives

$$\begin{aligned}\log(W_1/W_0) &= b\sqrt{M} \left[\sqrt{\varphi+1} - \sqrt{\varphi} - \frac{1}{2} \right] + \log M + \log(4\sqrt{3}) - 1 \\ &\quad - \log((\varphi+1)/\varphi) + O(M^{-1/2}).\end{aligned}\tag{74}$$

To verify the constant rather than infer it from a fit, decompose the ratio into $-\log p(M)$, the linear-edge difference, and the residual-partition difference. These are, respectively,

$$\begin{aligned}&-b\sqrt{M} + \log M + \log(4\sqrt{3}) + O(M^{-1/2}), \\ M[(\varphi+1)f'(M-1/2) - \varphi f'(M+1/2)] &= b\sqrt{M}/2 - 1 + O(M^{-1/2}), \\ f((\varphi+1)M) - f(\varphi M) &= b\sqrt{M}[\sqrt{\varphi+1} - \sqrt{\varphi}] - \log((\varphi+1)/\varphi) + O(M^{-1/2}).\end{aligned}$$

Their sum is (74) uniformly on a fixed neighborhood of $9/16$.

Let $D(\varphi) = \sqrt{\varphi+1} - \sqrt{\varphi} - 1/2$. Then $D(9/16) = 0$ and $D'(9/16) = -4/15$. The smooth equal-leading-weight center has the expansion

$$\varphi_*(M) = \frac{9}{16} + \frac{15}{4b\sqrt{M}} \left[\log M + \log \frac{36\sqrt{3}}{25} - 1 \right] + O\left(\frac{\log^2 M}{M}\right).\tag{75}$$

The positive shift converges slowly. The width is $M^{-1/2}$. With

$$u = \frac{4b}{15} \sqrt{M}(\varphi - \varphi_*(M)),$$

where the center is defined by the exact smooth leading-carrier equality, $W_1/W_0 \rightarrow e^{-u}$ and the upper-charge proportion tends to $(1 + e^{-u})^{-1}$. The common C_0 cancels. First relative corrections change the center only at order M^{-1} . [write] Indeed, the first relative corrections change $\log(W_1/W_0)$ by $B_1(c_{M-1})/\sqrt{M-1} - B_1(c_M)/\sqrt{M} = O(M^{-1/2})$, while by (74) its derivative in φ near 9/16 is $b\sqrt{M} D'(\varphi) + O(1) = -(4b/15 + o(1))\sqrt{M}$; the centre therefore moves by $O(M^{-1/2})/\sqrt{M} = O(M^{-1})$ (added in the write).

The limiting fraction 9/16 is not an accurate finite- M threshold by itself. Uncertified binary64 calculations give equal-leading-weight phases approximately 1.02188178 at $M = 1000$, 0.71601628 at 10^4 , 0.58296425 at 10^6 , and 0.56518714 at 10^8 . The first lies outside the physical cell. These are diagnostics of the slow approach, not rigorous enclosures or onset bounds.

11.2 An explicit expansion of the exact normalization

Expanding (3) gives

$$f(k) = b\sqrt{k} - \log k + c_0 + d_1 k^{-1/2} + d_2 k^{-1} + O(k^{-3/2}),$$

where $c_0 = -\log(4\sqrt{3})$, $d_1 = -b/48 - 1/b$, and $d_2 = 1/24 - 1/(2b^2)$. Euler–Maclaurin summation, with the exponentially small integer error already controlled, gives

$$\begin{aligned} \log P_M &= \frac{2b}{3} M^{3/2} - M \log M + (1 + c_0)M + \left(\frac{11b}{24} - \frac{2}{b}\right) \sqrt{M} \\ &\quad + \left(-\frac{11}{24} - \frac{1}{2b^2}\right) \log M + C_P + O(M^{-1/2}), \end{aligned} \quad (76)$$

with the absolutely convergent explicit constant

$$\begin{aligned} C_P &= \sum_{k \geq 1} \left[\log p(k) - b\sqrt{k} + \log k - c_0 - \frac{d_1}{\sqrt{k}} - \frac{d_2}{k} \right] \\ &\quad + b\zeta(-1/2) + d_1\zeta(1/2) + d_2\gamma - \frac{1}{2} \log(2\pi). \end{aligned} \quad (77)$$

The summand is $O(k^{-3/2})$. Subtracting more differentiated asymptotic terms accelerates the constant and generates every further half-power coefficient by classical Euler–Maclaurin machinery. For example the next coefficient, multiplying $M^{-1/2}$, is

$$\frac{73b}{2304} - \frac{11}{24b} + \frac{2}{3b^3}.$$

It follows from $d_3 = -b/4608 - 1/(48b) - 1/(3b^3)$ through $b/24 + d_1/2 - 2d_3$. No certified numerical value of C_P is claimed in the package. Keeping P_m exact avoids a normalization constant error entirely. [write] A certified value of C_P is question (ii) of Section 15; no theorem of this Part needs it.

11.3 Proof of the discrete inverse theorem

Fix J and let $x = x_J(Y)$. By (9), at integer k in the relevant large range,

$$|\log a(k) - \log \mathcal{A}_J(k)| \leq C_J M(k)^{-(J+1)/2}.$$

The positive slope (10) makes the sufficiently large carrier strictly increasing and unbounded, hence its root unique. Choose a strict margin $\Delta = D_J M(x)^{-J/2}$ larger than the coefficient error divided

by the lower slope, uniformly in the small surrounding interval. Every integer below $\lceil x - \Delta \rceil$ lies strictly below $x - \Delta$ and therefore has $\log a(k) < Y$. The integer $\lceil x + \Delta \rceil$ lies at or above $x + \Delta$ and has $\log a(k) \geq Y$. Monotonicity proves (14); the same mean-value estimate proves the sequence-value assertion. This uses only pointwise errors at integers. Differentiating an unknown coefficient remainder or postulating a true real crossing is unnecessary.

12 Joint charge and low hole laws

Under the probability measure on size- n objects, outer selections have weights proportional to $\prod_{k \in S} p(k)$. With $L = \lfloor M^{17/20} \rfloor$, define

$$\mathcal{H}_n = [1, L] \setminus S, \quad Q_n = |S| + |\mathcal{H}_n|, \quad K_n = |S| = Q_n - |\mathcal{H}_n|.$$

The effective filled-edge charge Q_n is different from the actual outer count K_n . Define the moving probability

$$\pi_1(n) = \frac{W_{M-1}(n)}{W_M(n) + W_{M-1}(n)},$$

and let B_n be Bernoulli of parameter $\pi_1(n)$, independent of $\mathcal{H}_\infty$.

Theorem 12.1 (Uniform positive mass approximation). *Embedding $\mathcal{H}_n$ into finite subsets of $\mathbb{N}$, uniformly through crossovers and triangular boundaries,*

$$d_{\text{TV}}(\text{Law}(Q_n - M, \mathcal{H}_n), \text{Law}(-B_n, \mathcal{H}_\infty)) = O(M^{-1/2}). \quad (78)$$

Put

$$\mu_{\text{count}} = \sum_{k \geq 1} \frac{1}{p(k) + 1}, \quad \sigma_{\text{count}}^2 = \sum_{k \geq 1} \frac{p(k)}{(p(k) + 1)^2}.$$

Then

$$\mathbb{E}K_n = M - \pi_1(n) - \mu_{\text{count}} + O(M^{-1/2}), \quad (79)$$

$$\text{Var } K_n = \pi_1(n)(1 - \pi_1(n)) + \sigma_{\text{count}}^2 + O(M^{-1/2}). \quad (80)$$

This is a moving comparison measure, so no fixed phase subsequence is required. The actual count never reduces asymptotically to only two values: it includes the nondegenerate low-hole shift.

Proof. The exact unnormalized joint mass at $(m, \mathcal{H})$ is $p(\mathcal{H})^{-1}$ times the charge- m filled coefficient at $n + h$. The two-stage weighted truncations, the global localization, and the carrier-ratio bound reduce to $h \leq M^\epsilon$ with superpolynomial lost joint mass relative to $a(n)$. Positivity is crucial: the sum of charge-resolved discarded masses equals the nonnegative global discarded mass, so the unmarked localization estimate controls their total absolute mass.

For the dominant charges with $r_m/m \geq 1/4$, the local theorem gives their filled mass at $n + h$ as $W_m(n + h)(1 + O(M^{-1/2}))$. Before using a derivative, replace both integer values $p(r_m)$ and $p(r_m + h)$ by their exponentially accurate smooth values $e^{f(r_m)}$, $e^{f(r_m + h)}$. Apply (64) to these explicit smooth carriers. For integer $h \geq 1$ the replacement errors are absorbed, giving

$$\left| \frac{W_m(n + h)}{W_m(n)} - 1 \right| \leq ChM^{-1/2}e^{Ch/\sqrt{M}}. \quad (81)$$

At $h = 0$ the ratio is exactly one, handled separately. No interpolation or derivative of $p(r)$ is asserted. The small-residual charge is exponentially subdominant. If $n + h$ crosses a triangular boundary,

its extra charge has residual $o(M)$ and is likewise negligible, while the common residual-near-one charge remains dominant.

Thus the raw L^1 error in replacing the exact joint mass by $p(\mathcal{H})^{-1}W_m(n)$ for $m = M, M-1$ is at most

$$CM^{-1/2}(W_M + W_{M-1}) \sum_{\mathcal{H} \subset [1, L]} \frac{(1+h)e^{Ch/\sqrt{M}}}{p(\mathcal{H})}.$$

The sum is uniformly bounded by (69). Extending the comparison set to all finite holes costs only a superpolynomial error. Its total mass is $C_0(W_M + W_{M-1})$. Dividing by the two normalizing totals changes the error only by a fixed factor, proving (78).

Total variation alone would not prove unbounded moments. Multiply this raw L^1 estimate separately by $(1+h)^j$ for $j = 1, 2$, and use the corresponding fixed tilted moments. On the retained two charges, $|K_n - M| \leq 1+h$. On discarded exact tails, $K_n \leq M$, $Q_n \leq M+L$, and $h \leq L(L+1)/2$, so fixed polynomial weights are absorbed by the superpolynomial tail estimates. The comparison law has finite stretched-exponential energy moments. Hence the first two centered moments of $K_n - M$ differ by $O(M^{-1/2})$. Independence of B_n and the limiting holes gives (79)–(80). $\square$

13 Exact checks and certified hole constants

The proof is analytic; finite agreement does not certify an asymptotic remainder or its onset. The accompanying standard-library code independently checks the finite algebra and normalizations, with integer or rational arithmetic throughout the exact path.

13.1 Independent recurrences and combinatorial tests

Ordinary partition numbers through 10000 are computed by unbounded coin-change dynamic programming. A separate recurrence checks them through 600:

$$np(n) = \sum_{j=1}^n \sigma_1(j)p(n-j).$$

The descending 0/1 product dynamic program computes $a(n)$ through 5000. An algebraically different recurrence follows by logarithmic differentiation of (1). If

$$d(j) = \sum_{k|j} k(-1)^{j/k+1} p(k)^{j/k},$$

then

$$na(n) = \sum_{j=1}^n d(j)a(n-j).$$

Every numerator is checked for exact divisibility by n and agreement through 600. The initial values are

$$1, 1, 2, 5, 8, 18, 34, 65, 109, 223, 386, 698, 1241, 2180, 3804, 6788.$$

The original source check agreed with all 37 displayed initial values in the inspected live OEIS entry. The attempted b-file fetch failed; no full OEIS b-file comparison is claimed.

An independent enumeration covers all 81156 ordinary partitions of sizes 0 through 35. It uses doubled half-integer coordinates to check the row and Frobenius formulas, conjugation for $Q_1, \dots, Q_6$, and both exact q-brackets (39). The low-hole identity is independently exhausted for

$0 \leq n \leq 28$ and $0 \leq L \leq 5$: 174 cases and 8898 triples. Checks include energy, unique recovery, exact missing set, actual/filled cardinality, integer weight cancellation, Frobenius weight equality, and every changed denominator exceeding L .

These are independent finite checks of implementations and algebra. The explicit inverse map and analytic estimates, not enumeration, prove the general assertions.

13.2 A pure integer interval certificate

For $g(j) = j(j+3)/2 \leq k$, every subset of $\{2, \dots, j+1\}$ can be completed by ones to a distinct partition of k , so $p(k) \geq 2^j$. The block $g(j) \leq k < g(j+1)$ has $j+2$ integers. Let $T_s(N) = \sum_{k>N} k^s/p(k)$. With $N = 10000$, $j_0 = 139$, $g(j_0) = 9869$, and 132 already-included integers in that block, exact geometric power sums yield

$$T_0(N) \leq \frac{2j_0 + 6 - 132}{2^{j_0}} = \frac{19}{87112285931760246646623899502532662132736}, \quad (82)$$

$$\begin{aligned} T_1(N) &\leq \frac{j_0^3 + 10j_0^2 + 37j_0 + 56 - 132g(j_0 + 1)}{2^{j_0}} \\ &= \frac{390677}{174224571863520493293247799005065324265472}. \end{aligned} \quad (83)$$

For T_1 we bound every k in a block by $g(j+1)$; summing $(j+2)g(j+1)2^{-j}$ and subtracting the corresponding first-block majorant gives the displayed polynomial. The identities use $\sum_{i \geq 0} i^r 2^{-i} = 2, 2, 6, 26$ for $r = 0, 1, 2, 3$.

All finite products and sums are enclosed by outward-rounded integer arithmetic on the scale 10^{100} . For example, if $[l, u]/10^{100}$ encloses a product, multiplying by $(p(k) + 1)/p(k)$ updates the lower endpoint by floor division and the upper by ceiling division. Sums use the same directed divisions. The tail multiplier for C_0 is at most $e^{T_0} \leq 1/(1 - T_0)$, so no transcendental interval routine is needed. The remaining tails for mean energy, mean count and count variance are at most T_1, T_0, T_0 respectively.

Rounding the resulting exact intervals further outward gives the following shorter readable enclosures:

$$\begin{aligned} 7.60150293364724365227288154074110141857 &< C_0, \\ C_0 &< 7.60150293364724365227288154074110141858, \\ 6.08634257165234578177405930993613318243 &< \mu_1, \\ \mu_1 &< 6.08634257165234578177405930993613318468, \\ 1.69334813525351189239046960428498655116 &< \mu_{\text{count}}, \\ \mu_{\text{count}} &< 1.69334813525351189239046960428498655117, \\ 1.21130990884242092190904749282953762059 &< \sigma_{\text{count}}^2, \\ \sigma_{\text{count}}^2 &< 1.21130990884242092190904749282953762060. \end{aligned}$$

The package contains the full 100-decimal rational endpoints. They lie strictly inside an independently produced earlier interval certificate. This comparison itself is performed by exact rational inequalities.

13.3 Floating diagnostics and their limits

Optional floating files are separate from the exact core and certificates. They test carrier ratios, slow crossover locations, and product-normalization trends. Selected leading and first-corrected ratios to the exact coefficient are:

n	Leading carrier divided by $a(n)$	First corrected divided by $a(n)$
100	0.1495141	0.9306268
400	0.00669010	0.03071625
1000	0.000346150	0.00122480
2000	0.00124365	0.00410422
3200	0.000954154	0.00281057
5000	0.00177838	0.00486243

These are uncertified numerical evaluations of the eventual formula at small indices. In this range slow $\log M/\sqrt{M}$ centering and additional preasymptotic charges matter. The table supplies no effective threshold beyond which the formula is accurate. A Gaussian’s close fit here is likewise not a proof of its eventual coefficient behavior.

The all-orders construction in this report is effective through Bloch–Okounkov brackets and finite moments, but the package does not implement a general symbolic all-orders engine. It implements the exact finite core, stated first-normalization checks, rigorous hole intervals, and optional diagnostics. Nor does it implement an interval-certified finite- n coefficient approximation from the asymptotic carrier.

13.4 Reproducible packaging and trust boundaries

The source package includes the TeX report, Python exact checker, frozen exact vectors and reference certificates, separate optional diagnostics, a builder, guard tests, and a manifest verifier. Explicit runtime guards remain active under `python -O`. The deterministic builder uses an isolated Python invocation, disables TeX shell escape, rejects unresolved references and overfull boxes, and creates a sorted ZIP with fixed metadata and an exact file inventory. Extracted rebuilding is checked for byte identity with the same installed Python/TeX stack; identity across different toolchain versions is not promised.

The manifest detects accidental changes; it is not a signature protecting against coordinated replacement of both files and manifest. The build assumes trusted source, Python/TeX installations and executable search path. Disabling shell escape does not turn TeX into a sandbox for arbitrary hostile inputs. No network or external upload is required to run the package. The README gives precise commands and the final build receipt records hashes separately from the self-contained archive.

[write] (5 October 2026.) In this repository the package’s files are shipped under the prefix `182-strict-` (code in `code/`, data in `data/`); the delivery README is replaced by the report README, the manifest `SHA256SUMS.json` and the PDF are not shipped, and the 1.43 MB table `data/references/coefficients_0_5000.txt`, which `code/verify.py` reads, is not shipped either. The commands work only in the delivered layout; the report README gives the retrieval from the arrival commit and rerun routes on copies.

14 Attribution and the limits of the source screen

The exact sequence, product and twice-partitioned interpretation are attributed to OEIS A271619 and its contributor Gus Wiseman [3]. Hardy–Ramanujan/Rademacher estimates and eta identities are classical [4]; shifted-symmetric quasimodularity and the effective n -point construction are Bloch–Okounkov’s [5]; modified-Bessel transfer expansions are classical [6]. Generic all-orders saddle-point transfer, including Harris–Schoenfeld admissibility as discussed by Odlyzko–Richmond, is established

methodology [12]. The target-specific work here is verification of localization, tame remainder bounds, finite hole assembly and their consequences for this amplitude product.

The bounded comparison screen inspected the following particularly relevant theorem scopes:

- Granovsky–Stark’s multiplicative framework has factors $S(a_k z^k)^{b_k}$ with $0 < a_k \leq 1$ in its stated theorem [7]. Substituting $a_k = p(k)$ in the target factor $1 + a_k z^k$ violates that amplitude bound. Related “sub exponential” papers retain this restriction [8, 9]; that phrase in their titles does not license arbitrary growing amplitudes.
- Expansive selection theorems concern $\prod(1 + q^k)^{a_k}$ [10]. They select distinct component types and therefore count a different class.
- Bridges–Brindle–Bringmann–Franke provide genuine arbitrary-order expansions for $\prod(1 - q^n)^{-f(n)}$ under nonnegative multiplicities and Dirichlet-series hypotheses [11]. A formal Euler-product rewrite does not establish those hypotheses for the present amplitudes.
- Fermi–Dirac integral expansions and physical Fermi level-counting/shell effects are relevant prior methods and phenomena [13, 14, 15, 16]. No originality is claimed for Fermi smoothing or discrete oscillations. The full theorem text of Garoni–Frankel–Glasser and the full Leboeuf paper were not retrieved in the screen; they are credited as background, not used as an exhaustive theorem-level exclusion.

The actual complete prior A022629 article was also checked within the project [1]. Its main family covers power amplitudes, fixed convolution powers and integer slot multiplicities; its section on more general weights does not supply a theorem for these stretched amplitudes. Prior A291698 moving-fugacity material [2] concerns a different model and explicitly leaves its beyond-polynomial regime outside its result. Exact-ID/title repository and Library searches produced no direct match, but negative searches are not proofs of absence.

[write] (5 October 2026, batch 100.) Checked at the write: the passages of [1] and [2] described here (question 7 of the A022629 report; Section 10.3 of the A291698 report) are as described; neither report has changed since 2 October 2026, before this manuscript’s date. No other report in the repository treats A271619. Part II of this report treats the neighbouring sequence A358836, and Section 1 records the relations to these reports and to the new report on A126348.

Our precise conclusion is only that no inspected primary theorem directly preempted the target result in this bounded screen. We make no universal novelty claim. A bibliographic lead to Gingold–Quaintance’s work on integer power products remained unverified at full-theorem level and is not used to support either coverage or noncoverage.

Review record. Separate mathematical reviews reconstructed the local remainder/transfer proof, the global Bellman/entropy proof, and the full finite-moment assembly and inverse interface. The optional total-variation proof was reviewed after clarifying that exact integer partition weights are smoothed before differentiation and that the zero-shift ratio is treated exactly. An independent finite checker recomputed the algebra and constant intervals. A preliminary working-dossier resonance moment incorrectly omitted the upper-region restriction; the manuscript’s split in Section 6 corrects that before publication. No finite computation is presented as validation of an unproved analytic gate.

[write] (5 October 2026.) The reviews named here were made within the delivering session; their records are not part of the delivery, apart from the provenance files of the exact checks (`data/182-strict-PROVENANCE.json`, `data/182-strict-SECOND_ORDER_PROVENANCE.json` and

the frozen independent checkers). No referee or reviewer outside that session has checked the proofs, and the repository adds no review of its own beyond the notes marked [write].

15 Further questions and research

[write] Section title changed in the write (5 October 2026, batch 100) from “Further directions”. The first list is the manuscript’s own. The second, added in the write under ProveIt’s standing rule of 4 October 2026, collects the claims of this Part that are stated without proof, with the source, its sketch and what is missing.

The following are proposed directions for additional work, not assertions that each is globally open in the literature.

- Make the asymptotic onset effective, or develop a practical positive multi-charge approximation with finite-index error bounds. The poor moderate-index performance documented here makes this a substantive numerical problem.
- Determine how far the fixed-order estimates can be made uniform in a growing truncation order, and whether optimal truncation exposes controlled exponentially small corrections. No such conclusion follows from the present all-fixed-orders expansion.
- Prove coefficient-level phase theorems for more general stretched amplitudes, including the putative $\alpha = 1/3$ transition suggested by the resonance modulus. That observation alone supplies no coefficient theorem.
- Extend the positive joint-law argument to further conditioned diagram statistics, and investigate higher-order distributional corrections while keeping the persistent low-hole cloud separate from edge fluctuations.

Claims of Part I stated without proof ([write]).

- (i) *The stretched-amplitude modulus law.* Report 182, Section 6, “A bounded general observation”: if $\log w(k) \sim bk^\alpha$ with $0 < \alpha < 1$, “with sufficiently controlled differentiated remainders and negligible lower-edge contributions”, then $\log |\phi(2\pi\ell/X)| \sim -2\pi^4\ell^2 X^{1-3\alpha}/(3b^3(1-\alpha)^3)$, so that $\alpha = 1/3$ is a natural modulus threshold. The source calls it an observation and asserts no coefficient theorem. Sketch: the computation of (20), with $t \sim bX^{\alpha-1}$ and $H_t \sim X^{1-\alpha}/(b(1-\alpha))$. Missing: the hypotheses made precise (which remainder control, which bound on the lower region, whose proof for $\alpha = 1/2$ uses $v_k \leq Ce^{-c\sqrt{k}}$), a proof under them, and any coefficient-level consequence (the manuscript’s third direction above). The write notes a parallel: at $\alpha = 1/3$ the resonance damping is of order one, as at $\log u \asymp n^{1/6}$ in Section 10.3 of the A291698 report and in the critical theta window of Report 93 (Section 22 of the A022629 report, batch 100); whether a theta-type multiplier over many charges replaces the two-charge phase there is open.
- (ii) *A certified value of C_P .* Section 11 gives C_P as the absolutely convergent series (77) and claims no certified value. Missing: an explicit bound for the $O(k^{-3/2})$ summand (an explicit-error form of (4)) and interval values of $\zeta(-1/2)$, $\zeta(1/2)$ and γ . No theorem of this Part uses C_P : P_m is kept exact.

Nothing else in Part I was found stated without proof. The one-line claim that first relative corrections move the crossover centre only at order M^{-1} is justified in a note in Section 11.

Part II

Nested partitions with distinct block lengths (A358836)

[write] Part II is Report 181 of the session bundle, *Nested Partitions with Distinct Block Lengths: All fixed asymptotic orders, explicit relative growth, inverse bounds and conditioned limits for OEIS A358836*, author line “Research report 181”, dated 3 October 2026 (21 pages as delivered). Its abstract follows. Its text is unchanged except for the label prefix `stp:len:`, the citation key of its OEIS entry, the title of Section 27 (was “Further questions”) and the notes marked [write]; its running headers and table-of-contents spacing were not carried over. Its letters differ from Part I’s (Table 1): here b is a variance, $a = \pi^4/72$, $\mu(z) = \log P(e^{-z})$ and E_h are Edgeworth corrections.

Abstract of Part II. Let a_n count multisets of integer partitions of total weight n in which the numbers of parts of the constituent blocks are distinct. Andrew Howroyd’s recorded product for OEIS A358836 is $\prod_{k \geq 1} (1 + q^k / \prod_{j \leq k} (1 - q^j))$. We reduce its logarithm to a moving-argument Jacobi product, the ordinary partition product, the MacMahon product and a finite-end correction. The latter admits a computable formal expansion with uniform analytic remainders on disks of radius proportional to t^2 , where $q = e^{-t}$.

An all-angle characteristic-function bound gives an exact-saddle coefficient expansion through every fixed Edgeworth order, with a rigorously integrated remainder. The leading logarithm grows as $n^{3/4}$, followed by explicit $n^{1/2} \log n$ and $n^{1/4}$ power-log terms. We also give the constant-order completion, yielding a fully elementary relative equivalent with error $O(n^{-1/4} \log^4 n)$. Controlled inversion includes a bounded-error numerical carrier and arbitrary-order two-floor envelopes; neither an unconditional rounding rule nor arbitrary accuracy against a fixed piecewise-linear interpolation is asserted.

For uniform objects of weight n , the number of occupied lengths has a Gaussian limit, and the largest occupied length has a Gumbel limit, jointly independent. The moving complex saddle and truncated-product conditioning steps are proved. The product and composition interpretation are prior, and the modular, Mellin, saddle-point and conditioning methods are classical. The report establishes sequence-specific deductions and records a bounded source comparison, without a global priority claim. Exact finite checks and optional floating-point diagnostics are kept separate.

16 The model and the scope of the results

A block is an ordinary integer partition. Its *length* is its number of positive parts, and its weight is their sum. An object is an unordered collection of blocks, with no two blocks of the same length. The total weight is the sum of the block weights. The empty collection has weight zero. Write a_n for the number of objects of weight n .

This is the sequence A358836 [17], whose initial values are

$$1, 1, 2, 4, 8, 15, 28, 51, 92, 164, 289, 504, 871, 1493, 2539, 4290, 7201, 12017, 19939$$

The restriction concerns block *lengths*, not block weights or distinct block contents. This distinction separates the model from A271619 and A261049 [18].

Define

$$P_k(q) = \prod_{j=1}^k (1 - q^j)^{-1}, \quad B_k(q) = q^k P_k(q), \quad F(q) = \sum_{n \geq 0} a_n q^n = \prod_{k \geq 1} (1 + B_k(q)). \quad (84)$$

Conjugation of partitions shows that B_k counts partitions with exactly k parts. Taking zero or one block of each length gives the last identity. This is the elementary specialization of Howroyd’s

formula recorded on 31 December 2022; it is not a new product. For $|q| \leq r < 1$, $|B_k(q)| \leq r^k P(r)$, where

$$P(q) = \prod_{j \geq 1} (1 - q^j)^{-1}.$$

Thus the product converges normally in $|q| < 1$ and defines an analytic function there.

Gus Wiseman's composition interpretation, recorded on 21 August 2024, has a direct bijection. Conjugate each block, sort the resulting partitions by their distinct largest parts in increasing order, and concatenate them. Inside each block the parts weakly decrease. Between consecutive blocks there is an ascent, since the old last part is at most its leader and the new leader is larger. Hence the blocks are exactly the maximal weakly decreasing runs, whose leaders strictly increase. Conversely, split such a composition into these runs and conjugate them.

Appending a part 1 preserves this property and is injective. For $n \geq 1$ the one-part composition $(n + 1)$ is not in its image. Consequently

$$a_{n+1} > a_n \quad (n \geq 1), \quad 1 \leq a_n \leq 2^{n-1} \quad (n \geq 1). \quad (85)$$

The repeated initial value $a_0 = a_1 = 1$ causes no difficulty for eventual inversion.

Throughout, put

$$A = \frac{\pi^2}{6}, \quad Z = \zeta(3), \quad a = \frac{A^2}{2}, \quad s_* = (3a)^{1/4} = \frac{\pi}{24^{1/4}}, \quad L(t) = \log F(e^{-t}). \quad (86)$$

An expansion through *all fixed orders* means that the order is fixed before $t \downarrow 0$ or $n \rightarrow \infty$. The error constants may depend on that order. We do not assert convergence of the complete formal series, bounds uniform in a growing truncation order, an effective asymptotic onset or numerical interval certificates for the asymptotic errors.

The Jacobi, Dedekind eta and MacMahon identities are classical [19, 20, 21, 22]. The complete MacMahon logarithmic expansion is part of the Ramanujan–Wright theory; the inspected proof is Koshkin's [23], with Wright [24] cited historically. Saddle-point and Gaussian-moment coefficient extraction are standard [25]. The model-specific work below is the nested-denominator reduction, its small-disk estimates and the global coefficient and conditioning arguments. Section 26 states exactly the limits of the source comparison.

17 An exact reduction to classical products

Let $q = e^{-z}$, $\operatorname{Re} z > 0$, and use the logarithms defined by convergent power series to set

$$\mu_k(z) = \log P_k(e^{-z}), \quad \mu(z) = \log P(e^{-z}), \quad M(q) = \prod_{j \geq 1} (1 - q^j)^{-j}.$$

Define

$$\begin{aligned} D(z) &= \prod_{k \geq 1} (1 + e^{\mu(z) - kz}), & T_k(z) &= \mu(z) - \mu_k(z), \\ u_k(z) &= e^{kz - \mu_k(z)}, & w_k(z) &= e^{\mu(z) - kz}. \end{aligned}$$

For positive real z , and then wherever the quotients are analytic, the elementary factor identity is

$$\frac{1 + B_k(e^{-z})}{1 + P(e^{-z})e^{-kz}} = \frac{P_k(e^{-z})}{P(e^{-z})} \frac{1 + u_k(z)}{1 + w_k(z)^{-1}}.$$

Moreover,

$$\prod_{k \geq 1} \frac{P_k(q)}{P(q)} = \prod_{j \geq 1} (1 - q^j)^{j-1} = \frac{P(q)}{M(q)}.$$

All extracted products converge absolutely on the positive axis. Thus

$$F(e^{-z}) = D(z) \frac{P(e^{-z})}{M(e^{-z})} R(z), \quad R(z) = \prod_{k \geq 1} R_k(z), \quad (87)$$

where the useful unsplit factor is

$$R_k(z) = \frac{w_k(z) + e^{T_k(z)}}{w_k(z) + 1} = 1 + \frac{e^{T_k(z)} - 1}{1 + w_k(z)}. \quad (88)$$

For large k one must estimate this quotient as a whole. Separating its numerator and denominator logarithms into two infinite sums would introduce artificial divergence.

17.1 The analytic disk and its denominators

Fix a sufficiently small constant $c > 0$ and work on

$$\mathcal{D}_t = \{z : |z - t| \leq ct^2\}, \quad t \downarrow 0. \quad (89)$$

Slightly smaller concentric disks will be used for Cauchy estimates. A disk with radius proportional to t is not justified: edge factors can have zeros at distances of order t^2 .

Lemma 17.1. *Uniformly for every $k \in \mathbb{N} \cup \{\infty\}$ and $z \in \mathcal{D}_t$,*

$$\mu_k(z) - \mu_k(t) = O(c).$$

There is a fixed $\eta > 0$ such that, for all k and sufficiently small t ,

$$|1 + w_k(z)| \geq \eta \max(1, |w_k(z)|). \quad (90)$$

The same conclusion holds with $\mu(z)$ replaced by $\mu(z) + s$ for all complex s in a sufficiently small fixed disk.

Proof. For $\operatorname{Re} z \asymp t$,

$$|\mu'_k(z)| \leq \sum_{j \geq 1} \frac{j}{e^{j \operatorname{Re} z} - 1} = O(t^{-2}),$$

independently of k . Integrating along the segment from t proves the first assertion. The eta transformation [21] gives uniformly on $\mathcal{D}_t$

$$\mu(z) = \frac{A}{z} + \frac{1}{2} \log \frac{z}{2\pi} - \frac{z}{24} + O(e^{-d/t}), \quad (91)$$

with the logarithm continued from the positive axis and some $d > 0$. For example the exact identity has residual product at $Q = e^{-4\pi^2/z}$, whose logarithm is exponentially small on this disk.

If $|w_k| \leq 1/2$ or $|w_k| \geq 2$, the reverse triangle inequality proves (90). Otherwise (91) implies

$$k = \frac{A}{t^2} + O\left(\frac{|\log t|}{t}\right), \quad |\operatorname{Im}(\mu(z) - kz)| \leq 2Ac + o(1).$$

Choose c so that this is less than $\pi/4$. In this transition band w_k stays in a fixed sector about the positive axis, proving the bound. Reserving some slack in the sector proves the assertion with $\mu + s$. $\square$

17.2 The finite-end expansion

For fixed J , define the rational numbers h_j formally by

$$H(z) = \sum_{k \geq 1} \log \left(1 + e^{kz} \prod_{j=1}^k (1 - e^{-jz}) \right) = \sum_{j \geq 1} h_j z^j. \quad (92)$$

This is a formal power series definition: the summand indexed by k has order z^k , so the coefficient of z^j uses only $k \leq j$. The displayed infinite logarithmic sum is not claimed to converge for real $z > 0$.

Proposition 17.2. *For every fixed integer $J \geq 1$, R has a zero-free analytic logarithm on $\mathcal{D}_t$ for all sufficiently small t , and*

$$\log R(z) = \sum_{j=1}^J h_j z^j + O_J(t^{J+1}) \quad (93)$$

uniformly there. Its first ten coefficients are

$$\begin{aligned} h_1 &= 1, & h_2 &= 2, & h_3 &= 7, & h_4 &= \frac{68}{3}, & h_5 &= \frac{391}{4}, & h_6 &= \frac{40043}{90}, \\ h_7 &= \frac{96787}{40}, & h_8 &= \frac{17366357}{1260}, & h_9 &= \frac{157752233}{1728}, & h_{10} &= \frac{285439229213}{453600}. \end{aligned} \quad (94)$$

Proof. Choose a small fixed $\delta > 0$ and a large fixed $C > 0$. We estimate four ranges, using Lemma 17.1 throughout.

For $k \leq \delta/t$, the inequality $|1 - e^{-jz}| \leq j|z|$, valid for $\operatorname{Re} z \geq 0$ by integration along a segment, gives

$$\left| \frac{u_{k+1}(z)}{u_k(z)} \right| = |e^z(1 - e^{-(k+1)z})| \leq r < 1.$$

Since $u_1 = e^z - 1 = O(t)$ and $u_{J+1} = O_J(t^{J+1})$, the sum of $|\log(1 + u_k)|$ over $J < k \leq \delta/t$ is $O_J(t^{J+1})$.

For $\delta/t \leq k \leq C/t$, the first $\lfloor \delta/t \rfloor$ partition factors on the real axis give $\mu_k(t) \geq d_\delta/t$ with $d_\delta > 0$. This follows, for instance, from a positive lower Riemann sum for $-\log(1 - e^{-x})$ on a fixed subinterval of $(0, \delta)$. The uniform $O(c)$ transfer and $k \operatorname{Re} z \leq C + o(1)$ then give

$$|u_k(z)| \leq e^{-d_\delta/(2t)}.$$

For $C/t \leq k \leq A/(2t^2)$, use

$$\mu(t) - \mu_k(t) = O(t^{-1}e^{-kt}) \quad (kt \geq C).$$

Choosing C large and using (91) yields $\operatorname{Re} \mu_k(z) \geq 3A/(4t)$. It follows that $|u_k(z)| \leq e^{-A/(5t)}$ after decreasing the constant in the exponent. These two ranges contain $O(t^{-2})$ indices. Their total is therefore exponentially small. On all $k \leq A/(2t^2)$, the complementary terms $\log(1 + e^{kz - \mu(z)})$ also have exponentially small total.

Finally, for $k \geq A/(2t^2)$,

$$|T_k(z)| \leq C_1 t^{-1} e^{-k \operatorname{Re} z}.$$

Equation (88) and (90) show that $|\log R_k| \leq C_2 |T_k|$ on this range. Consequently

$$\sum_{k \geq A/(2t^2)} |\log R_k(z)| = O\left(t^{-2} e^{-A/(2t) + O(1)}\right).$$

All these logarithms can be chosen analytically and close to zero; the finitely many remaining factors $1 + u_k$ are also close to one. This proves zero-freeness of R and

$$\log R(z) = \sum_{k=1}^J \log(1 + u_k(z)) + O_J(t^{J+1}).$$

Taylor expansion of the fixed finite sum proves (93). Rational formal arithmetic gives (94). $\square$

18 The complete free-energy expansion

The reduction (87) isolates all the large terms. We give the classical normalizations explicitly because a shift error in the theta square would change the lower-order coefficients.

18.1 Jacobi normalization on the small disk

The triple product [19] gives the exact identity

$$D(z)(-e^{-\mu(z)}; e^{-z})_{\infty}(e^{-z}; e^{-z})_{\infty} = \sum_{m \in \mathbb{Z}} \exp\{-\mu(z)m - zm(m-1)/2\}. \quad (95)$$

Put $\gamma = 1/2 - \mu(z)/z$. Completing the square produces the prefactor $\exp((\mu - z/2)^2/(2z))$. Gaussian Poisson summation [20] gives

$$\sum_{m \in \mathbb{Z}} e^{-z(m-\gamma)^2/2} = \sqrt{\frac{2\pi}{z}} \sum_{l \in \mathbb{Z}} e^{-2\pi^2 l^2/z} e^{2\pi i l \gamma},$$

where the square root is continued from the positive axis. On $\mathcal{D}_t$,

$$\operatorname{Re}(1/z) = t^{-1} + O(c), \quad |\operatorname{Im}(\mu(z)/z)| \leq \frac{2Ac + o(1)}{t}.$$

Hence a nonzero dual term is bounded by

$$\exp \left\{ \frac{-2\pi^2 l^2 + (4\pi Ac + o(1))|l|}{t} + O(cl^2) \right\}.$$

Decreasing c makes the sum of all these terms $O(e^{-d/t})$. Also $\log(-e^{-\mu}; e^{-z})_{\infty} = O(t^{-1}e^{-\operatorname{Re}\mu})$. Taking analytic logarithms of (95), and then using (91), gives

$$\begin{aligned} \log D(z) &= \frac{\mu^2}{2z} - \frac{\mu}{2} + \frac{z}{8} + \frac{1}{2} \log \frac{2\pi}{z} + \mu + O(e^{-d/t}) \\ &= \frac{\mu^2}{2z} + \frac{A}{z} - \frac{\mu}{2} + \frac{z}{12} + O(e^{-d/t}). \end{aligned} \quad (96)$$

These bounds also establish the required local analytic logarithm of D .

[write] (5 October 2026, batch 100.) With $u = e^{\mu(z)}$, $D(z)$ is the product $\prod_{k \geq 1} (1 + uq^k)$ of the A291698 report [32], whose exact Jacobi decomposition (its equation (11)) continues analytically to complex u ; its logarithm, $(\mu^2/2 + A)/z + z/12 - \mu/2$ plus the logarithms of the dual theta sum and of $(Q; Q)_{\infty}^{-1}$ and $(-1/u; e^{-z})_{\infty}^{-1}$, is the second line of (96). The identity is classical and is derived here independently; what this Part adds is the bound on the dual terms on the moving complex disk (89). Identity (95) is the instance $q = e^{-z}$, $z \mapsto -e^{-\mu(z)}$ of Jacobi's triple product as formalized in Lean (`Fabius.hasSum_jacobi_triple_product`; Section 1).

18.2 MacMahon residues

Mellin inversion gives, initially on $\operatorname{Re} w > 2$,

$$\log M(e^{-z}) = \frac{1}{2\pi i} \int \Gamma(w) \zeta(w+1) \zeta(w-1) z^{-w} dw. \quad (97)$$

This follows by expanding $-\log(1-e^{-jz})$ and using the Mellin transform of the exponential; absolute convergence justifies interchange. The pole at $w = 2$ contributes Z/z^2 . The double pole at zero contributes $(\log z)/12 + \zeta'(-1)$, and the poles at $w = -2m$ contribute

$$c_{2m} = \frac{\zeta(1-2m)\zeta(-1-2m)}{(2m)!}, \quad c_2 = -\frac{1}{2880}. \quad (98)$$

These are the classical Ramanujan–Wright coefficients [23, 24]. Gamma decay dominates the polynomial zeta bounds in any fixed closed sector $|\arg z| \leq \alpha < \pi/2$. Shifting the contour past sufficiently many poles therefore gives, for each fixed J ,

$$\log M(e^{-z}) = \frac{Z}{z^2} + \frac{1}{12} \log z + \zeta'(-1) + \sum_{1 \leq m \leq J/2} c_{2m} z^{2m} + O_J(t^{J+1}) \quad (99)$$

uniformly on $\mathcal{D}_t$. If a desired remainder exponent coincides with a pole, shift farther, include that residue and absorb terms of the target size into the remainder. This does not assert convergence of the full residue series.

Theorem 18.1. *Write $\ell = \log(t/(2\pi))$. Then*

$$\begin{aligned} L(t) = & \frac{a}{t^3} + \frac{A\ell/2 - Z}{t^2} + \frac{\ell^2/8 + 35A/24}{t} \\ & + \frac{7}{48} \log t - \frac{11}{48} \log(2\pi) - \zeta'(-1) + \frac{1225}{1152} t + \frac{5761}{2880} t^2 + 7t^3 + O(t^4). \end{aligned} \quad (100)$$

At every fixed higher order the expansion is obtained by substituting (91), (93), (96) and (99) into

$$L(z) = \frac{\mu(z)^2}{2z} + \frac{A}{z} + \frac{\mu(z)}{2} + \frac{z}{12} - \log M(e^{-z}) + \log R(z) + O(e^{-d/t}). \quad (101)$$

Every fixed number of derivatives has the corresponding differentiated expansion with natural remainder order. In particular, for each fixed $r \geq 0$,

$$(-1)^r L^{(r)}(t) \sim a(3)_r t^{-3-r}, \quad |L^{(r)}(z)| = O_r(t^{-3-r}) \quad (102)$$

on a smaller fixed disk, where $(3)_r = 3 \cdot 4 \cdots (r+2)$ and $(3)_0 = 1$.

Proof. The product identity gives $L = \log D + \mu - \log M + \log R$. Substitution proves (101) and elementary collection of terms gives (100). For example the uncombined regular term from the first three classical terms is $73t/1152$, to which $h_1 t = t$ is added; the t^2 term is $h_2 - c_2 = 5761/2880$.

The derivative statement needs care. Cauchy's estimate on a radius- ct^2 disk loses *two* powers of t for each derivative. To differentiate a truncation through t^J a fixed r times with remainder $O(t^{J+1-r})$, first establish the undifferentiated expansion through t^{J+r} . Cauchy's estimate for its remainder gives

$$O(t^{J+r+1-2r}) = O(t^{J+1-r}).$$

The derivatives of the extra explicitly retained powers $t^{J+1}, \dots, t^{J+r}$ are of the same or smaller order. Nested disks give the analogous uniform bounds near t . The leading term is at^{-3} , proving (102). $\square$

19 The exact saddle and a global arc bound

Under the Boltzmann law with parameter e^{-t} , an object of weight N has probability proportional to e^{-tN} . The block choices are independent. Block k is absent with probability $(1 + B_k)^{-1}$; conditional on its presence its weight is

$$k + \sum_{j \leq k} jG_j, \quad \mathbb{P}(G_j = r) = (1 - e^{-tj})e^{-tjr} \quad (r \geq 0). \quad (103)$$

In particular $-L'(t)$ is the mean total weight and $L''(t) > 0$ is its variance. The mean decreases continuously from infinity to zero as t increases from zero to infinity. Thus for each $n > 0$ there is exactly one $t = t_n > 0$ such that

$$-L'(t_n) = n. \quad (104)$$

By (102),

$$t_n \sim s_* n^{-1/4}, \quad b = L''(t_n) \sim 12at_n^{-5} = 6A^2t_n^{-5}. \quad (105)$$

Lemma 19.1. *For sufficiently small real $t > 0$, there is a constant $c_1 > 0$ such that, for every $-\pi \leq \theta \leq \pi$,*

$$\frac{|F(e^{-t+i\theta})|}{F(e^{-t})} \leq \exp\{-c_1 \min(\theta^2/t^5, t^{-3})\}. \quad (106)$$

Proof. Fix $0 < \alpha < \beta < A$ and select all integers $k \in [\alpha/t^2, \beta/t^2]$. There are $\Theta(t^{-2})$ such blocks. For them

$$B_k(e^{-t}) = \exp\{(A - kt^2)/t + O(|\log t|)\},$$

so their absence probabilities ε_k satisfy $\varepsilon_k \leq e^{-d/t}$ for some $d > 0$.

Each selected occupied block contains all the geometric variables with $j \in [1/t, 2/t]$. For $r_j = e^{-tj}$, uniformly bounded away from zero and one,

$$\left| \frac{1 - r_j}{1 - r_j e^{ij\theta}} \right|^2 = \left(1 + \frac{4r_j \sin^2(j\theta/2)}{(1 - r_j)^2} \right)^{-1} \leq e^{-c \sin^2(j\theta/2)}.$$

The remaining conditional factors have modulus at most one. Hence the modulus of the occupied-block characteristic function is at most

$$\exp \left\{ -c \sum_{1/t \leq j \leq 2/t} \sin^2(j\theta/2) \right\} \leq \exp\{-c_2 \min(\theta^2/t^3, 1/t)\}. \quad (107)$$

To see the second inequality on the full interval, use three ranges. For $|\theta|/t$ sufficiently small, a sine lower bound gives a constant times $\theta^2 \sum j^2$. For $|\theta|/t$ in a fixed positive compact interval, the rescaled sum converges uniformly to the positive integral $\int_1^2 \sin^2(x\theta/(2t)) dx$. For $|\theta| \geq Ct$ with C large, use

$$\sum \sin^2(j\theta/2) = \frac{\#\{j\}}{2} - \frac{1}{2} \operatorname{Re} \sum e^{ij\theta}, \quad \left| \sum e^{ij\theta} \right| \leq \frac{1}{|\sin(\theta/2)|}.$$

Together these cover all angles, including roots of unity.

It remains to retain attenuation after mixing in the absent state. Put $u = \min(\theta^2/t^3, 1/t)$ and decrease c_2 to $c_0 \leq \min(c_2, d/2)$. For any $\varepsilon \leq e^{-d/t}$,

$$\begin{aligned} \log\{\varepsilon + (1 - \varepsilon)e^{-c_0 u}\} &= -c_0 u + \log\{1 + \varepsilon(e^{c_0 u} - 1)\} \\ &\leq -c_0 u + \varepsilon c_0 u e^{c_0 u} \\ &\leq -c_0 u(1 - e^{-d/(2t)}) \leq -c_0 u/2. \end{aligned}$$

We used $e^x - 1 \leq xe^x$ and $u \leq 1/t$. Multiplying over the $\Theta(t^{-2})$ selected blocks gives (106); every unused normalized factor has modulus at most one. $\square$

Cauchy's formula at the exact saddle reads

$$a_n = \frac{e^{nt+L(t)}}{2\pi} \int_{-\pi}^{\pi} \frac{F(e^{-t+i\theta})}{F(e^{-t})} e^{-in\theta} d\theta. \quad (108)$$

Only on a zero-free local arc do we express the ratio by an exponential involving $L(t - i\theta)$. No global logarithm across possible zeros is required. Lemma 19.1 supplies coefficient control that a positive-real asymptotic alone would not provide.

20 Every fixed Edgeworth order

Set, at $t = t_n$,

$$\kappa_j = (-1)^j L^{(j)}(t), \quad b = \kappa_2.$$

For $h \geq 1$, define

$$E_h(t) = \sum_{\substack{m_3, \dots, m_{2h+2} \geq 0 \\ \sum (j-2)m_j = 2h}} \frac{(-1)^{J/2} (J-1)!!}{b^{J/2}} \prod_{j=3}^{2h+2} \frac{\kappa_j^{m_j}}{m_j! (j!)^{m_j}}, \quad J = \sum_{j=3}^{2h+2} j m_j, \quad (109)$$

and put $E_0 = 1$. The letter J in this formula is the total Gaussian degree, not a free-energy truncation order. Its parity is even.

Theorem 20.1. *For every fixed integer $H \geq 0$,*

$$a_n = \frac{\exp(nt_n + L(t_n))}{\sqrt{2\pi L''(t_n)}} \left\{ \sum_{h=0}^H E_h(t_n) + O_H(t_n^{3H+3}) \right\}, \quad E_h(t) = O(t^{3h}). \quad (110)$$

The first two corrections are

$$E_1 = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} \sim -\frac{35}{4\pi^4} t^3, \quad (111)$$

$$E_2 = -\frac{\kappa_6}{48b^3} + \frac{7\kappa_3\kappa_5}{48b^4} + \frac{35\kappa_4^2}{384b^4} - \frac{35\kappa_3^2\kappa_4}{64b^5} + \frac{385\kappa_3^4}{1152b^6}. \quad (112)$$

Proof. Fix $0 < \delta < 1/2$ and take the core $|\theta| \leq t^{5/2-\delta}$. Its complement has exponentially small integral by (106). [write] At $|\theta| = t^{5/2-\delta}$ the bound (106) is $\exp(-c_1 t^{-2\delta})$: superpolynomially small in t , not exponentially small in $1/t$. This suffices for every fixed order (correction of wording, 5 October 2026). The core lies in the analytic disk (89) for small t .

Put $v = \theta\sqrt{b}$ and $\rho = t^{3/2}$. By (102), the standardized cumulants

$$\lambda_j = \frac{\kappa_j}{b^{j/2}} = O(\rho^{j-2}) \quad (j \geq 3)$$

have fixed-order bounds. Taylor expansion through degree $2H+3$ gives

$$L(t - i\theta) - L(t) - in\theta = -\frac{v^2}{2} + \sum_{j=3}^{2H+3} \lambda_j \frac{(iv)^j}{j!} + R_H, \quad (113)$$

$$|R_H| \leq C_H \rho^{2H+2} |v|^{2H+4}.$$

The derivative bound in Theorem 18.1, obtained by expanding extra orders before differentiation, is what gives this remainder. The total non-Gaussian exponent, including R_H , is $o(v^2)$ uniformly on the core: its leading ratio is $O(\rho|v|) = o(1)$.

Assign weight $j - 2$ to $\lambda_j(iv)^j/j!$. Expand its exponential, retaining all products of total weight at most $2H + 1$. Here is an explicit integrable remainder argument. Introduce an auxiliary variable r , replace each term of weight d by r^d times that term, and apply Taylor's theorem at $r = 0$ through degree $2H + 1$, evaluated at $r = 1$. Each monomial in the remainder derivatives has total weight at least $2H + 2$. Since only finitely many cumulant terms occur, these derivatives are bounded by

$$C_H \rho^{2H+2} P_H(|v|) e^{\epsilon_t v^2}, \quad \epsilon_t \rightarrow 0,$$

for a fixed polynomial P_H . The same form bounds the effect of R_H in (113). Multiplication by $e^{-v^2/2}$ gives an integrable Gaussian majorant and total error $O_H(\rho^{2H+2}) = O_H(t^{3H+3})$. Gaussian tails allow the retained polynomial integrals to extend from the core to the whole real axis.

The parity of $\sum j m_j$ equals the parity of $\sum (j - 2) m_j$, so all odd weights integrate to zero. For an even total degree J ,

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-v^2/2} (iv)^J dv = (-1)^{J/2} (J - 1)!!.$$

The even weight $2h$ therefore gives exactly (109). The prefactor from (108) is $1/\sqrt{2\pi b}$. This proves (110) with no logarithmic loss in the integrated remainder. Direct tuple enumeration gives (111) and (112); substitution of $\kappa_j \sim a(3)_j t^{-3-j}$ gives the stated leading constant. $\square$

21 Explicit growth and an elementary relative equivalent

Put

$$\begin{aligned} \varepsilon &= n^{-1/4}, & \ell_n &= \log \frac{s_* \varepsilon}{2\pi}, & B_n &= \frac{A}{2} \ell_n - Z, \\ C_n &= \frac{\ell_n^2}{8} + \frac{35A}{24}, & D_n &= B_n - \frac{A}{4}. \end{aligned} \tag{114}$$

Theorem 21.1. *The first saddle correction is*

$$t_n = s_* \varepsilon + \frac{D_n}{2s_*^2} \varepsilon^2 + O(\varepsilon^3 (\log n)^2). \tag{115}$$

The three divergent power-log tiers are

$$\log a_n = \frac{4s_*}{3} n^{3/4} + \frac{B_n}{s_*^2} n^{1/2} + \left(\frac{C_n}{s_*} - \frac{D_n^2}{2s_*^5} \right) n^{1/4} + O((\log n)^3). \tag{116}$$

In particular,

$$\log a_n = \frac{4\pi}{3 \cdot 24^{1/4}} n^{3/4} - \frac{\sqrt{6}}{24} n^{1/2} \log n + O(n^{1/2}). \tag{117}$$

There is a recursive power-log expansion of $\log a_n$ to any prescribed fixed algebraic accuracy, obtained by the substitution described below.

Proof. Write $t = \varepsilon U$ in the saddle equation. Then

$$1 + \varepsilon^4 L'(\varepsilon U) = 0.$$

Its leading term is $1 - 3a/U^4$, which vanishes at $U = s_*$ and has derivative $4/s_*$ there. Recursively solve with

$$U = s_* + \sum_{j \geq 1} \varepsilon^j u_j(\log \varepsilon).$$

At each stage the coefficient of the next unknown polynomial is $4/s_*$, so the recursion is unique. The first one is $u_1 = D_n/(2s_*^2)$.

This is more than a formal construction. The differentiated expansion in Theorem 18.1 gives a controlled residual after any fixed finite substitution. The derivative of the scaled equation remains bounded away from zero near s_* . The mean value theorem, or the local analytic implicit-function theorem with the same bounds, turns the residual into a corresponding error for the true saddle. Thus the recursion gives genuine asymptotic expansions with polynomial logarithms.

Substitute in

$$nt + L(t) - \frac{1}{2} \log(2\pi L''(t)) + \log \left(\sum_{h=0}^H E_h(t) \right). \quad (118)$$

The first-order saddle displacement cancels by stationarity. The second-order contribution is the negative completed square $-D_n^2/(2s_*^5)$ in (116). The Gaussian logarithm and Edgeworth terms are below the three displayed tiers. Keeping more free-energy, saddle and Edgeworth terms in (118) gives any fixed power accuracy. One may over-truncate by a fixed amount, avoiding any claim about optimal truncation. Exponentiating a remainder tending to zero gives the corresponding relative expansion. $\square$

The three tiers alone are not a relative equivalent because their omitted logarithmic terms still diverge. The next result completes all terms down through order one.

Theorem 21.2. *Define*

$$G_n = \frac{D_n^3}{3s_*^8} - \frac{AD_n^2}{8s_*^8} + \frac{D_n(\ell_n/4 - C_n)}{2s_*^4} \quad (119)$$

and the fully explicit expression

$$\begin{aligned} \mathcal{T}(n) = & \frac{4s_*}{3} n^{3/4} + \frac{B_n}{s_*^2} n^{1/2} + \left(\frac{C_n}{s_*} - \frac{D_n^2}{2s_*^5} \right) n^{1/4} + G_n \\ & + \frac{7}{48} \log(s_* \varepsilon) - \frac{11}{48} \log(2\pi) - \zeta'(-1) + \frac{5}{2} \log \varepsilon - \frac{1}{2} \log \frac{8\pi}{s_*}. \end{aligned} \quad (120)$$

Then

$$\log a_n = \mathcal{T}(n) + O(n^{-1/4}(\log n)^4), \quad a_n = e^{\mathcal{T}(n)} \{1 + O(n^{-1/4}(\log n)^4)\}. \quad (121)$$

Proof. The first three exponent functions in the variable $U = t/\varepsilon$ are

$$\begin{aligned} \phi_0(U) &= U + a/U^3, \\ \phi_1(U) &= \{A(\ell_n + \log(U/s_*))/2 - Z\}/U^2, \\ \phi_2(U) &= \{(\ell_n + \log(U/s_*))^2/8 + 35A/24\}/U. \end{aligned}$$

Thus $nt + L(t)$ begins with $\varepsilon^{-3}\phi_0(U) + \varepsilon^{-2}\phi_1(U) + \varepsilon^{-1}\phi_2(U)$. At $U = s_*$,

$$\phi_0'' = \frac{4}{s_*}, \quad \phi_0''' = -\frac{20}{s_*^2}, \quad \phi_1'' = \frac{6B_n - 5A/2}{s_*^4}, \quad \phi_2' = \frac{\ell_n/4 - C_n}{s_*^2}.$$

With $u_1 = D_n/(2s_*^2)$, the constant Legendre contribution is

$$\frac{1}{6}\phi_0'''u_1^3 + \frac{1}{2}\phi_1''u_1^2 + \phi_2'u_1 = G_n.$$

Any terms involving the next saddle coefficient u_2 cancel because $\phi_0''u_1 + \phi_1' = 0$. The logarithmic part of (100) gives the next three terms of (120). Finally

$$b = \frac{4}{s_*}\varepsilon^{-5}\{1 + O(\varepsilon|\log \varepsilon|)\}$$

gives the last two Gaussian terms there.

For the error, set $t_0 = s_*\varepsilon + u_1\varepsilon^2$. Equation (115) gives $t_n - t_0 = O(\varepsilon^3(\log n)^2)$. By stationarity, replacing t_n by t_0 in $nt + L(t)$ costs

$$O(\varepsilon^{-5}(t_n - t_0)^2) = O(\varepsilon(\log n)^4).$$

The fourth-order Taylor term of ϕ_0 , the third-order one of ϕ_1 and the second-order one of ϕ_2 are of this order or smaller. Changes in the logarithmic term, Gaussian prefactor, first regular t term and Edgeworth corrections are smaller. This proves the first assertion in (121). Its remainder tends to zero, so exponentiation proves the second. $\square$

22 Controlled inversion and its rounding limits

For $x \geq 1$, let

$$N(x) = \max\{n \geq 0 : a_n \leq x\}, \quad Y = \log x.$$

Equation (85) and the growth theorem make this finite and eventually unambiguous. Put $c = 4s_*/3$ and $N_0 = (Y/c)^{4/3}$.

Proposition 22.1. *As $x \rightarrow \infty$,*

$$N(x) = N_0 - \frac{N_0^{3/4}}{s_*^3} \left\{ \frac{A}{2} \log \frac{s_* N_0^{-1/4}}{2\pi} - Z \right\} + O(N_0^{1/2}(\log N_0)^2). \quad (122)$$

Proof. The derivative of the leading carrier $cn^{3/4}$ is $s_*n^{-1/4}$. Dividing the second growth tier $B_n n^{1/2}/s_*^2$ by this slope gives the displayed displacement. Taylor expansion of the leading carrier and the next tier after this displacement leaves vertical error $O(N_0^{1/4}(\log N_0)^2)$, which gives the stated horizontal remainder. Rounding contributes at most a bounded amount and is smaller. $\square$

For a much sharper carrier, let $L_2(t)$ denote precisely the right side of (100) through its t^2 term, with no remainder, and set

$$Q_2(t) = L_2(t) - tL_2'(t) - \frac{1}{2}\log(2\pi L_2''(t)). \quad (123)$$

For small positive t , $L_2''(t) > 0$ and

$$Q_2'(t) = -tL_2''(t) - \frac{L_2'''(t)}{2L_2''(t)} \sim -6A^2t^{-4} < 0.$$

Thus the small positive solution of $Q_2(\tau) = Y$ is unique for all sufficiently large Y . Define

$$\nu_2(Y) = -L_2'(\tau), \quad Q_2(\tau) = Y. \quad (124)$$

This is an explicitly specified one-dimensional root problem involving only the displayed constants and elementary functions.

Theorem 22.2. *Let I be the strictly increasing piecewise-linear interpolant with $I(n) = \log a_n$ for integers $n \geq 1$. Then*

$$I^{-1}(Y) - \nu_2(Y) = O(Y^{-2/3}), \quad N(x) = \nu_2(\log x) + O(1). \quad (125)$$

No fixed floor of $\nu_2(Y)$ is asserted to equal $N(e^Y)$ for every sufficiently large Y .

Proof. The differentiated free-energy theorem gives

$$L - L_2 = O(t^3), \quad (L - L_2)' = O(t^2), \quad (L - L_2)'' = O(t).$$

At the exact saddle t_n , Theorem 20.1 with $H = 0$ and these bounds give

$$\log a_n - Q_2(t_n) = O(t_n^3), \quad -L_2'(t_n) - n = O(t_n^2).$$

If $Q_2(\tau) = \log a_n$, then $\tau - t_n = O(t_n^7)$ since $|Q_2'| \asymp t_n^{-4}$. Multiplication by $L_2'' = O(t_n^{-5})$ shows

$$\nu_2(\log a_n) - n = O(t_n^2).$$

For a statement between integers, define $g_2(v) = Q_2(t_2(v))$, where $-L_2'(t_2(v)) = v$. Direct differentiation gives $g_2'(v) = t_2(v)(1 + o(1))$ and $g_2''(v) = O(t_2(v)^5)$. Its linear interpolation on a unit interval has vertical error $O(t^5)$, while its endpoint errors relative to $\log a_n$ are $O(t^3)$. Dividing by a slope asymptotic to t gives the uniform horizontal error $O(t^2)$. Since $Y \asymp t^{-3}$, this is $O(Y^{-2/3})$.

For Y in the interval $[\log a_n, \log a_{n+1})$, $N(e^Y) = n = \lfloor I^{-1}(Y) \rfloor$; the same identity holds at the endpoints. This proves the bounded-error assertion, but a small smooth error can still change a floor arbitrarily near an integer. $\square$

22.1 Higher-order inverse information

It is important not to strengthen (125) incorrectly. A smooth carrier for $\log a_n$ has curvature of order t^5 on a unit interval. Its piecewise-linear interpolation therefore contributes a vertical error of order t^5 , or a horizontal error of order t^4 , away from the endpoints. Higher smooth asymptotic truncations alone cannot supply arbitrary accuracy relative to the *fixed* interpolant I^{-1} ; fractional-part corrections would be needed.

The appropriate arbitrary-order discrete statement uses pointwise errors and two floors.

Theorem 22.3. *Let $g_R(v)$ be a sufficiently high fixed-order smooth saddle carrier, obtained from (118), increasing for large real v . Suppose, for fixed $p > 1$ and $m \geq 0$, its integer-point error has the proved bound*

$$|\log a_n - g_R(n)| \leq C t_n^p (1 + |\log t_n|)^m.$$

Write $g_R(\nu_R(Y)) = Y$, and let t be the saddle scale at $\nu_R(Y)$. For a sufficiently large fixed C' put

$$\eta_R(Y) = C' t^{p-1} (1 + |\log t|)^m.$$

Then for all sufficiently large Y ,

$$\lfloor \nu_R(Y) - \eta_R(Y) \rfloor \leq N(e^Y) \leq \lfloor \nu_R(Y) + \eta_R(Y) \rfloor. \quad (126)$$

The exponent p can be made arbitrarily large by adding enough free-energy and Edgeworth terms.

Proof. The construction and differentiated remainders give $g'_R(v) \sim t(v)$, uniformly over the small relevant horizontal range. With a strict margin in C' , the change in g_R from ν_R to either boundary dominates the pointwise integer error. Thus the integer at or below the lower boundary has logarithmic coefficient at most Y , whereas every integer strictly above the upper boundary has logarithmic coefficient greater than Y . Monotonicity (85) then proves (126). Arbitrarily large fixed p follows from Theorems 20.1 and 21.1. $\square$

If both floors agree, the threshold is determined. If the carrier is exceptionally close to an integer, the envelope may retain two adjacent integers. This is a genuine rounding ambiguity, not a defect repaired by asserting an unproved exact floor rule.

23 Marking the number of occupied lengths

Let K_n be the number of blocks in a uniformly chosen object of total weight n . Its marked generating function is

$$F(q, e^s) = \prod_{k \geq 1} (1 + e^s B_k(q)), \quad A_n(s) = [q^n] F(q, e^s), \quad L(z, s) = \log F(e^{-z}, e^s).$$

Thus $\mathbb{E}e^{sK_n} = A_n(s)/A_n(0)$. Below $t = t_n$ is the unmarked saddle. Write

$$m_B(t) = L_s(t, 0) = \sum_{k \geq 1} \frac{B_k(e^{-t})}{1 + B_k(e^{-t})}. \quad (127)$$

The real-marker estimates alone would not justify differentiating a coefficient asymptotic. We therefore establish the fixed complex-marker version first.

23.1 Uniform marked free energy

Use small fixed $|s| \leq s_0$ and $|z - t| \leq ct^2$, decreasing s_0 and c if needed. The denominator argument in Lemma 17.1 now uses $\mu + s$. The marked Jacobi factor is $D_s(z) = \prod_{k \geq 1} (1 + e^{\mu(z) + s - kz})$. In the dual theta sum from Section 18, replacing μ by $\mu + s$ adds at most $(2\pi s_0 + o(1))|l|/t$ to the exponent of the l th dual term. Shrinking c and s_0 therefore retains exponential suppression of all $l \neq 0$; the complementary product remains exponentially close to one. Consequently, uniformly on the joint complex disks,

$$\log D_s(z) = \frac{(\mu + s)^2}{2z} + \frac{A}{z} - \frac{\mu + s}{2} + \frac{z}{12} + O(e^{-d/t}).$$

In the finite-end factor replace u_k by $e^{-s}u_k$. Its formal coefficients $h_j(s)$ are finite polynomials in e^{-s} and hence analytic. The exact decomposition and the same four-region estimates give

$$L(z, s) - L(z, 0) = s \left(\frac{\mu(z)}{z} - \frac{1}{2} \right) + \frac{s^2}{2z} + \sum_{j=1}^J \{h_j(s) - h_j(0)\} z^j + O_J(t^{J+1}). \quad (128)$$

The remainder is uniform and analytic on smaller disks. Extra z orders before Cauchy differentiation supply natural mixed derivative remainders; Cauchy differentiation in s uses a fixed-radius disk. No analytic meaning is assigned to the divergent infinite formal H series.

In particular,

$$m_B(t) = \frac{\mu}{t} - \frac{1}{2} + O(t) = \frac{A}{t^2} + \frac{\log(t/(2\pi))}{2t} - \frac{13}{24} + O(t), \quad (129)$$

$$\begin{aligned} L_{ss}(t, 0) &= t^{-1} + O(t), \\ L_{ts}(t, 0) &= -2At^{-3} + O(t^{-2}|\log t|), \\ L_{tt}(t, 0) &= 6A^2t^{-5} + O(t^{-4}|\log t|). \end{aligned} \quad (130)$$

23.2 The moving complex saddle and global arcs

For $n = -L_t(t, 0)$ seek $\tau(s)$ satisfying $L_t(\tau(s), s) + n = 0$. Set $\tau = t + t^2v$. Uniformly for bounded v and small s ,

$$t^3\{L_t(t + t^2v, s) + n\} \longrightarrow 6A^2v - 2As.$$

The analytic implicit-function theorem applied to this scaled equation gives a unique analytic saddle on a sufficiently small fixed s disk, with

$$\tau(0) = t, \quad \tau(s) - t = O(|s|t^2), \quad \tau'(0) = \frac{t^2}{3A}(1 + o(1)). \quad (131)$$

Take the marker disk small enough that $\tau(s)$ stays in the zero-free z disk.

For the coefficient integral, choose the line $\operatorname{Re} z = \operatorname{Re} \tau$. Since n is an integer, its integrand is $2\pi i$ -periodic, so the integration interval can be centered at $\operatorname{Im} \tau$, giving $z = \tau + iy$, $-\pi \leq y \leq \pi$. This is a reparametrization of the same circle, with no extra contour-crossing hypothesis.

Choose a small fixed $\delta_0 > 0$ and then s_0 much smaller than δ_0 . On $|y| \leq \delta_0 t^2$, the uniform derivative expansion implies

$$\operatorname{Re} L_{tt}(\tau + iy, s) \geq c_3 t^{-5}.$$

Stationarity gives Gaussian decay $\exp(-c_4 y^2/t^5)$ throughout this band, including the intermediate local tails outside the Gaussian core. A positive-real modulus estimate centered at zero would not by itself establish this shifted local bound.

For the remaining arcs, put $r = \operatorname{Re} \tau$ and $\sigma = \operatorname{Re} s$. The proof of Lemma 19.1 applies to the marked factors normalized by $F(e^{-r}, e^\sigma)$: taking modulus and using the triangle inequality removes the phase of $e^{i \operatorname{Im} s}$ from the occupied term. Since $|\operatorname{Im} \tau| \leq Cs_0 t^2$, for $|y| \geq \delta_0 t^2$ the angular distance modulo 2π is at least $\delta_0 t^2/2$. The normalized bound is therefore $\exp(-c_5 \delta_0^2/t)$ or smaller.

To compare this with the complex saddle main term, expand from the real point (r, σ) to (τ, s) . The first-order change is purely imaginary. The second derivative bounds $L_{tt} = O(t^{-5})$, $L_{ts} = O(t^{-3})$, $L_{ss} = O(t^{-1})$ and $|\operatorname{Im} \tau| = O(t^2|\operatorname{Im} s|)$ yield

$$L(r, \sigma) - \operatorname{Re} L(\tau, s) = O(s_0^2/t).$$

Taking s_0 sufficiently small makes this cost strictly less than the negative global-arc exponent. Thus the global arcs are exponentially negligible relative to the complex saddle main term.

The weighted Taylor proof in Section 20 now applies uniformly for complex $|s| \leq s_0$. Set $b(s) = L_{tt}(\tau(s), s)$. It has positive real part and is asymptotic to $6A^2t^{-5}$; use the principal square root. The complex Gaussian moments have exactly the same formulas for $\operatorname{Re} b > 0$. In particular,

$$A_n(s) = \frac{e^{n\tau(s) + L(\tau(s), s)}}{\sqrt{2\pi b(s)}} \{1 + O(t^3)\}, \quad (132)$$

uniformly and analytically on a smaller fixed marker disk. The full fixed-order expansion also holds. Its uniform analytic remainder can be differentiated by Cauchy in s . [write] Only the first-order

form (132), with its $O(t^3)$ remainder and that remainder's Cauchy s -derivatives, is used below (Theorems 23.1 and 24.1). The full fixed-order marked expansion is asserted on the strength of the weighted Taylor proof applying uniformly; it is listed as question (vii) of Section 27 (5 October 2026).

23.3 The Gaussian block-count law

Theorem 23.1. *For uniform objects of weight n , with $t = t_n$,*

$$\mathbb{E}K_n = m_B(t) + O(t), \quad \text{Var } K_n \sim \frac{1}{3t}, \quad (133)$$

$$\frac{K_n - m_B(t)}{\sqrt{1/(3t)}} \implies \mathcal{N}(0, 1), \quad \frac{K_n}{\sqrt{n}} \longrightarrow \sqrt{\frac{2}{3}} \quad \text{in probability.} \quad (134)$$

More precisely,

$$\mathbb{E}K_n - m_B(t) = \frac{t}{3A} + O(t^2 |\log t|). \quad (135)$$

Proof. Take an analytic logarithm of (132) and write $\Psi_n(s) = \log A_n(s)$. Then

$$\Psi_n(s) = n\tau(s) + L(\tau(s), s) - \frac{1}{2} \log(2\pi b(s)) + O(t^3).$$

The saddle equation cancels the τ' terms in the first derivative of the optimized exponent, while $\tau' = -L_{ts}/L_{tt}$. Consequently

$$\Psi'_n(0) = m_B(t) - \frac{1}{2} \frac{b_s^{\text{total}}}{b} + O(t^3), \quad b_s^{\text{total}} = L_{tts} + L_{ttt}\tau'.$$

The leading terms are

$$L_{tts} \sim 6At^{-4}, \quad L_{ttt} \sim -30A^2t^{-6}, \quad \tau' \sim t^2/(3A),$$

so $b_s^{\text{total}} \sim -4At^{-4}$. The differentiated expansions sharpen this to (135).

[write] *Proof of (135), added in the write (5 October 2026).* Since $\mathbb{E}e^{sK_n} = A_n(s)/A_n(0)$, $\mathbb{E}K_n = \Psi'_n(0)$ exactly. The $O(t^3)$ remainder in (132) is analytic and uniform on a fixed marker disk, so its s -derivative at 0 is $O(t^3)$ by Cauchy's estimate; hence $\mathbb{E}K_n = m_B(t) - b_s^{\text{total}}/(2b) + O(t^3)$, all derivatives taken at $(t, 0)$. Each derivative involved has a leading term with relative error $O(t|\log t|)$: $L_{tt} = 6A^2t^{-5}(1 + O(t|\log t|))$ and $L_{ts} = -2At^{-3}(1 + O(t|\log t|))$ by (130); $L_{ttt} = -30A^2t^{-6}(1 + O(t|\log t|))$ by three differentiations of (100), legitimate by Theorem 18.1; and by (128) and (91), $L_s(z, 0) = A/z^2 + \log(z/2\pi)/(2z) - 13/24 + O(|z|)$, whose second z -derivative, legitimate by the mixed-derivative remainders stated after (128), is $L_{tts} = 6At^{-4}(1 + O(t|\log t|))$. Therefore $\tau'(0) = -L_{ts}/L_{tt} = \frac{t^2}{3A}(1 + O(t|\log t|))$, $b_s^{\text{total}} = L_{tts} + L_{ttt}\tau'(0) = -4At^{-4}(1 + O(t|\log t|))$ and

$$\mathbb{E}K_n - m_B(t) = -\frac{b_s^{\text{total}}}{2b} + O(t^3) = \frac{4At^{-4}}{12A^2t^{-5}}(1 + O(t|\log t|)) = \frac{t}{3A} + O(t^2 |\log t|).$$

[write] *A numerical comparison (binary64, uncertified).* Evaluating $L(t, s) = \sum_k \log(1 + e^s B_k(e^{-t}))$ and its derivatives termwise at the exact saddle reproduces the Boltzmann means $m_B(t)$ recorded in `data/181-nested-references-marked_verification.json` to ten digits, and $m_B(t) - b_s^{\text{total}}/(2b)$ agrees with the exact means $\mathbb{E}K_n$ recorded there to within $2.9 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$, $6.6 \cdot 10^{-4}$ and $4.1 \cdot 10^{-4}$ at $n = 100, 200, 400, 600$. The correction divided by t is 0.324, 0.332, 0.330, 0.324 there, still far from its limit $1/(3A) = 0.2026$; at $n = 600$, $\tau'(0)$ is 1.38 times $t^2/(3A)$. The first-order identity

is accurate already at these sizes; its leading constant is approached slowly. (The computation is a short script of the write, not shipped.)

[write] *What the proof rests on (5 October 2026)*. The proof of (135) is as strong as the uniform $O(t^3)$ remainder of (132) over complex markers, which the source asserts on the strength of its weighted Taylor proof. In this first-order form the remainder needs only bounds on L_{ttt} and L_{tttt} over the marker disk, and these follow from (128), with (100) at $s = 0$, by Cauchy estimates; the full fixed-order expansion of question (vii) is not used.

[write] *Dated note (5 October 2026)*. An independent adversarial check of this write, made by the intake after the write, found both of its completions valid: the proof of (135) above, and the marker-derivative bounds in the proof of Theorem 24.1. For the latter it noted that the complex- z form of the Bell-polynomial bound was used without its one-line justification, which is now supplied there. It computed a_n and $\mathbb{E}K_n$ independently, in exact integers from the bivariate product, for $n \leq 600$; they agree with the shipped terms 0 to 600 and with the shipped exact means. At the exact saddle, to 40 digits, it reproduced every figure of the comparison above. It followed the correction $-b_s^{\text{total}}/(2b)$, divided by t , to smaller t in binary64: 0.2441, 0.2233, 0.2114, 0.2072, 0.2051, 0.2037 and 0.2032 at $t = 0.1, 0.05, 0.02, 0.01, 0.005, 0.002$ and 0.001 , approaching $1/(3A) = 0.20264$, with its deviation from $1/(3A)$ divided by $t|\log t|$ bounded and decreasing. The check's programs are not shipped; it is not a formal verification or an external review.

The second derivative of the optimized exponent is the Schur complement

$$L_{ss} - \frac{L_{ts}^2}{L_{tt}} = \frac{1}{3t} + O(|\log t|).$$

Twice differentiating the logarithmic prefactor contributes a smaller order, giving the variance assertion. On a smaller fixed marker disk the third derivative of the optimized exponent is

$$L_{sss} + 3L_{tss}\tau' + 3L_{tts}(\tau')^2 + L_{ttt}(\tau')^3 = O(1).$$

This follows directly from (128): $L_{sss} = O(t)$, $L_{tss} = O(t^{-2})$, $L_{tts} = O(t^{-4})$, $L_{ttt} = O(t^{-6})$, and $\tau' = O(t^2)$. The prefactor and remainder also have bounded third derivatives. Taylor expansion at $s = iv\sqrt{3t}$, for fixed real v , gives the standard-normal characteristic function after centering by $m_B(t)$. The difference between this centering and the true mean is negligible on the standard-deviation scale.

Finally $m_B(t) \sim A/t^2$, $n \sim 3A^2/(2t^4)$ and the variance is lower order than n . These prove the last convergence in (134). $\square$

24 Largest length and joint independence

Let M_n be the largest occupied block length of a uniformly chosen object of positive weight n . This statistic concerns the number of parts in the largest-length block, not the largest part within a block. Define the standard Gumbel law by $\mathbb{P}(G \leq x) = \exp(-e^{-x})$.

Theorem 24.1. *With $t = t_n$ and $\mu(t) = \log P(e^{-t})$,*

$$tM_n - \mu(t) - \log(1/t) \implies G. \tag{136}$$

Moreover the pair

$$\left(\sqrt{3t}\{K_n - m_B(t)\}, tM_n - \mu(t) - \log(1/t)\right) \implies (Z_0, G), \tag{137}$$

where Z_0 is standard normal and Z_0 and G are independent.

Proof. Fix $x \in \mathbb{R}$ and put

$$m = \left\lfloor \frac{\mu(t) + \log(1/t) + x}{t} \right\rfloor.$$

For $k > m$, the partition tail estimate gives

$$P_k/P = 1 + O(t^{-1}e^{-kt}), \quad \max_{k>m} B_k = O(t), \quad \sum_{k>m} B_k^2 = O(t).$$

The relative partition-tail error is exponentially small here. Also

$$\sum_{k>m} P e^{-tk} = \frac{e^{\mu-t(m+1)}}{1 - e^{-t}} \longrightarrow e^{-x},$$

since the floor perturbs the exponent by only $O(t)$. Hence

$$T_m(z, s) = \sum_{k>m} \log(1 + e^s B_k(e^{-z})) \quad \text{satisfies} \quad T_m(t, 0) \longrightarrow e^{-x}. \quad (138)$$

The product with all $k > m$ deleted is $F_{\leq m} = F \exp(-T_m)$.

For every fixed $r \geq 1$,

$$\partial_z^r T_m(t, 0) = O(t^{-2r}). \quad (139)$$

Indeed, a conditional block in (103) has mean $k + O(t^{-2})$, and its higher geometric cumulant sums are $O(t^{-r-1})$. The Bell-polynomial expression for derivatives of B_k bounds them by a constant times $B_k(k + t^{-2})^r$. Summation against the exponentially decaying tail gives (139). The same estimates hold locally for $|z - t| \leq Ct^{5/2}$ and $|s| \leq C\sqrt{t}$, with the evident bounds for marker derivatives. [write] *Those bounds, added in the write (5 October 2026)*. Let $|z - t| \leq Ct^{5/2}$, $r = \operatorname{Re} z$, $|s| \leq C\sqrt{t}$ and $x_k = e^s B_k(e^{-z})$ for $k > m$. Then $|x_k| \leq e^{|s|} B_k(e^{-r}) \leq e^{|s|} P(e^{-r}) e^{-rk}$, and since $r - t = O(t^{5/2})$ changes μ by $O(\sqrt{t})$ and rm by $O(\sqrt{t})$ ($m = O(t^{-2})$), the computation before (138) gives $\sum_{k>m} |x_k| = O(1)$ and $\max_{k>m} |x_k| = O(t)$. The Bell-polynomial bound for the derivatives of B_k , stated above at real t , holds at such complex z with r in place of t (justification added 5 October 2026): $\log B_k(e^{-z}) = -kz - \sum_{j=1}^k \log(1 - e^{-jz})$, the p th z -derivative of $\log(1 - e^{-jz})$ is j^p times a polynomial without constant term in $1/(e^{jz} - 1)$, and $|e^{jz} - 1| \geq |e^{jz}| - 1 = e^{jr} - 1$ for $j \geq 1$, since $r > 0$ for small t . Hence $|\partial_z \log B_k(e^{-z})| \leq k + O(t^{-2})$ and $\partial_z^p \log B_k(e^{-z}) = O_p(t^{-p-1})$ for $p \geq 2$, so that $|\partial_z^p B_k(e^{-z})| \leq C_p |B_k(e^{-z})| (k + t^{-2})^p$. With $y = x/(1 + x)$, $\partial_s \log(1 + x) = y$ and $\partial_s y = y(1 - y)$, so every $\partial_s^r \log(1 + x_k)$ is a polynomial in y_k without constant term and is $O_r(|x_k|)$; similarly $\partial_z y = y(1 - y) \partial_z \log B_k$, and the Bell-polynomial bound above gives $|\partial_z^{r'} \partial_s^r \log(1 + x_k)| \leq C_{r,r'} |x_k| (k + t^{-2})^{r'}$. Summing against the geometric tail, $\partial_z^{r'} \partial_s^r T_m(z, s) = O_{r,r'}(t^{-2r'})$ on the whole region, and in particular $T_m(z, s) - T_m(z, 0) = O(|s|) = O(\sqrt{t})$.

The truncated product retains all the deep occupied lengths $k \in [\alpha/t^2, \beta/t^2]$ with $\beta < A$. It therefore has the same global suppression from Lemma 19.1. On every bounded Gaussian-scale interval $z = t + iy$, $y = O(t^{5/2})$, equation (139) gives

$$T_m(t + iy, 0) = T_m(t, 0) + O(\sqrt{t}).$$

Apply dominated convergence to the coefficient integrals, using the retained-block Gaussian bound for the tails. This proves

$$\frac{[q^n] F_{\leq m}(q)}{[q^n] F(q)} \longrightarrow \exp(-e^{-x}). \quad (140)$$

This is the required conditioning step. Boltzmann independence alone would not prove it. Equivalently, deleting the tail changes the mean weight by $O(t^{-2}) = o(\sqrt{b})$ and the variance by $O(t^{-4}) = o(b)$, consistently with the same local limit argument. Equation (140) proves (136).

For the joint law, take $s = iv\sqrt{3t}$ with v in a fixed real compact set. The complex saddle from (131) satisfies $\tau(s) - t = O(t^{5/2})$, so

$$T_m(\tau(s), s) = T_m(t, 0) + O(\sqrt{t}) \longrightarrow e^{-x}.$$

To make the contour domination explicit, let $r = \operatorname{Re} \tau(s)$. Since $r - t = O(t^{5/2})$, positivity and the geometric tail sum give

$$\sup_{\theta} \sum_{k>m} |e^s B_k(e^{-r-i\theta})| = O(1), \quad \sup_{k>m, \theta} |e^s B_k(e^{-r-i\theta})| = O(t).$$

Thus $|T_m| = O(1)$ on the entire contour. Multiplication by e^{-T_m} preserves the full marked integrand's Gaussian and outer-arc majorants up to a bounded factor. The local $O(\sqrt{t})$ tail perturbation above and dominated convergence now show

$$\frac{A_{n,\leq m}(s)}{A_n(s)} \longrightarrow \exp(-e^{-x}).$$

Multiplying by the centered block-count characteristic function gives

$$\mathbb{E} \left[e^{iv\sqrt{3t}(K_n - m_B(t))} \mathbf{1}_{\{tM_n - \mu(t) - \log(1/t) \leq x\}} \right] \longrightarrow e^{-v^2/2} e^{-e^{-x}}.$$

At every x , the limiting CDF is continuous. These mixed characteristic-function/CDF limits determine the product of the normal and Gumbel laws, proving (137). $\square$

25 Finite exact verification and diagnostic boundaries

The supplied source package separates a standard-library exact core from optional floating-point experiments. Finite verification checks the algebra and implementation; the proofs above establish the asymptotic conclusions.

25.1 Exact arithmetic checks

The core uses arbitrary-precision integers and rational numbers only. It computes $a_0, \dots, a_{600}$ by ordinary coefficient-array convolution, maintaining the partition polynomial P_k and multiplying successively by $1 + q^k P_k$. It agrees at every one of these 601 indices with the frozen coefficient data. A direct recursive composition enumeration, independent of the product computation, checks all weights $0 \leq n \leq 18$ using the increasing-run-leader characterization. The first 39 values of the supplied reference data were checked against the displayed OEIS entry during the source audit.

A finite rational power-series implementation computes all ten h_j in (94). Exact tuple enumeration checks both (111) and (112); substituting the leading cumulants checks the coefficient $-35/(144A^2)$ of t^3 in E_1 . Exact formal algebra checks the displayed free-energy coefficients through t^3 , the further formal positive-power coefficients through t^{10} , and the constant Legendre identity in Theorem 21.2. These checks do not assume convergence of H .

The composition bound in (85) can also justify a separate integer-encoding coefficient implementation: choosing a base larger than 2^{600} prevents digit carries for the finite range. Such an encoding is optional; the package's ordinary convolution provides the primary transparent computation.

25.2 Optional numerical experiments

The exploratory computation obtains cumulants directly from independent sums of geometric cumulants and Bernoulli/Bell-polynomial composition, rather than differentiating the claimed free-energy expansion. It compares exact coefficients with exact-saddle Gaussian and Edgeworth approximations at $n = 100, 200, 400, 600$. If

$$\mathcal{G}_n = \frac{e^{nt_n + L(t_n)}}{\sqrt{2\pi L''(t_n)}}, \quad r_H = \frac{a_n}{\mathcal{G}_n} - \sum_{h=0}^H E_h(t_n),$$

the exploratory values are

n	r_0	r_1	r_2
100	$-1.21515 \cdot 10^{-2}$	$-9.29042 \cdot 10^{-5}$	$-3.06935 \cdot 10^{-7}$
200	$-7.00214 \cdot 10^{-3}$	$-2.49941 \cdot 10^{-5}$	$4.21502 \cdot 10^{-7}$
400	$-3.95934 \cdot 10^{-3}$	$-6.05749 \cdot 10^{-6}$	$1.16867 \cdot 10^{-7}$
600	$-2.82485 \cdot 10^{-3}$	$-2.61937 \cdot 10^{-6}$	$4.89453 \cdot 10^{-8}$

These are rounded diagnostics from the supplied floating-point experiment, not interval enclosures, rigorous error certificates or evidence of a uniform error rate at these small indices. In particular the second corrected residual need not be monotone in this range.

Further optional experiments compare exact first and second block-count moments through $n = 600$, and exact largest-length cutoff coefficients at $n = 100, 200, 400, 600$, with the proved limiting carriers. Visible lattice effects and slow convergence are expected; they do not alter the fixed-order or weak-limit statements. No claims of optimized coefficient complexity or validated floating-point intervals are made.

25.3 Rebuilding the package

The editable source is `Report181.tex`. The source ZIP contains this manuscript, the exact core, frozen reference data, optional numerical scripts and their separate dependency information, a deterministic no-shell-escape builder, integrity checks and build tests. The README gives the precise commands. The standard build runs the exact checks and compiles with `pdflatex -no-shell-escape`; no optional numerical dependency is needed. A manifest records the packaged bytes. The PDF and ZIP are tested by extraction and rebuild. These reproducibility checks certify the finite supplied implementation and artifact identity, not the non-effective constants in asymptotic theorems.

[write] (5 October 2026.) In this repository the manuscript is Part II of `article.tex`; `Report181.tex` itself is not shipped, although `code/verify.py` and `guard_tests.py` read its marked OEIS prefix. The package's other files are shipped under the prefix `181-nested-`; its README, PDF, manifest and its copy of `verify_manifest.py` (identical to Part I's) are not. The commands work only in the delivered layout; the report README gives the retrieval from the arrival commit and rerun routes on copies.

26 Sources and the limits of the comparison

The primary OEIS record was read in full, including its formula, comments and references [17]. It credits the product to Howroyd and the composition interpretation to Wiseman; neither is presented here as a discovery. The inspected record links coefficient data through $n = 1000$ but does not give

an asymptotic formula or a research-paper citation. That observation concerns the inspected entry only.

Two related delivered reports were compared in the bounded audit. The first treats fixed norm-weighted products such as $\prod_k (1 + k^\alpha q^k)^\mu$ and multiplicity variants [31]; its full repository README and the first five pages of a related delivered PDF were read. The second treats $[q^n] \prod_k (1 + n^\alpha q^k)$ for fixed positive α [32]; its README and the relevant article text were read. Those reports already use all-order logarithmic and saddle expansions, inverse analysis and conditioned limit methods. Here the effective activity is $P(e^{-t})$, so $\log P(e^{-t}) \sim A/t \asymp n^{1/4}$ at the saddle, and it depends on the coefficient variable itself. It is therefore outside the directly substitutable polynomial-fugacity regime. This model distinction does not turn shared methods into new methods.

The inspected primary literature includes Brunel’s distinct-partition Fermi formulation [26], the open-access moving-argument q -product article of Arabi Ardehali and Rosengren [27], and the abstract and introductory definitions of Merca’s MacMahon-type series [29]. None of those inspected portions identifies the present coefficient problem. In particular, the uniform moving-argument range of Corollary 2 in [27] keeps a lower bound on the real part of its argument, whereas the effective argument here has real part of order $-A/t$. Direct substitution into that corollary is not justified. Its exact eta identity and the classical transformations remain relevant.

McIntosh’s complete q -shifted factorial expansion [28] was inspected only at the publisher abstract and metadata level; the subscription full text was not read. The unequal-block-size *labeled set partition* paper [30] concerns $\prod_k (1 + x^k/k!)$; only bibliographic material and a primary-paper search excerpt were available in this audit, and a direct full-text fetch failed. The full Kaneiwa text cited by neighboring A261049 was also inaccessible. The original Wright paper [24] was verified bibliographically, while the actual MacMahon theorem and proof were read in Koshkin [23]. None of these access limits is treated as a negative full-text finding.

Targeted searches for the sequence identifier, nested product, distinct block lengths, and run-leader interpretation did not locate an exact prior asymptotic match. Repository searches did not locate an A358836 duplicate. Library retrieval also had known misses for an already identified related report, so its negative results were incomplete. The audit was bounded, not exhaustive over journals, delivered artifacts or unpublished work. Accordingly this report makes no global novelty claim and does not characterize the results as a previously open-problem breakthrough. The classical ingredients are credited independently of the outcome of those searches. The companion source audit preserves the read-versus-inaccessible distinctions in more detail.

[write] (5 October 2026, batch 100.) The companion source audit is shipped as `181-nested-SOURCE_AUDIT.md`. The three repository files it read (the A022629 README and the A291698 README and article) are byte-identical at the write to the versions it records; no other report treats A358836. The A291698 report’s identity (11) is the one used for (96) (note there). Part I of this report treats the neighbouring sequence A271619, and the new report on A126348 (batch 100) treats another q -dependent fugacity, $1/(1 - q)$; Section 1 records these relations.

27 Further questions and research

[write] Section title changed in the write (5 October 2026, batch 100) from “Further questions”. Questions 1–6 are the manuscript’s own; question (vii) and the closing paragraph were added in the write under ProveIt’s standing rule of 4 October 2026.

The proofs leave several concrete directions open. They are questions suggested by this analysis, not claims that no prior literature addresses them.

1. Can one determine the large-order growth of the finite-end coefficients h_j and combine it

with the theta and MacMahon exponentially small terms to obtain an optimally truncated expansion? The present fixed-order estimates do not answer this.

2. Can the non-effective small- t constants be made explicit enough to give certified coefficient and inverse intervals at useful finite n ? This would require rigorous tail, arc and rounding controls in addition to the current diagnostics.
3. Does the rescaled collection of extreme occupied lengths converge to a Poisson point process with intensity $e^{-x} dx$, jointly with the Gaussian block count? The maximum theorem proves only the one-threshold statement and its joint independence.
4. What global shape and fluctuation field describe the occupied-length profile near its moving transition, after conditioning on total weight? The deep-occupied range and extreme tail treated here leave the transition window largely unexplored.
5. How do the asymptotics change if each length can occur up to a fixed multiplicity, or if the outer block restriction is modified while preserving nested partition denominators? The exact factorization would have to be rederived rather than assumed.
6. Can a fractional-part correction give an arbitrarily accurate explicit inverse of the particular piecewise-linear logarithmic interpolant? Section 22 identifies the t^4 horizontal curvature barrier that a smooth carrier alone cannot remove.

Claims of Part II argued only in outline ([write]).

- (vii) *The full fixed-order marked expansion.* Section 23, after (132): “The full fixed-order expansion also holds.” Sketch (source): the weighted Taylor proof of Theorem 20.1 applies uniformly for complex $|s| \leq s_0$, with $\operatorname{Re} b(s) > 0$ and the same complex Gaussian moments. Missing: complex-marker versions of the standardized cumulant bounds $\lambda_j = O(\rho^{j-2})$ and of the Taylor remainder (113), uniform on the marker disk (they should follow from the mixed-derivative remainders of (128)). Nothing in this Part uses more than the first-order form.

Two further steps given in outline are completed in the write and are not open: the sharpened mean (135) (“the differentiated expansions sharpen this”) and the marker-derivative bounds in the proof of Theorem 24.1 (“the evident bounds for marker derivatives”); both proofs are printed in place. Question 3 is the occupied-length counterpart of question 5 of the A022629 report [31] (the Poisson process of the largest parts there).

A Provenance, numbering and editorial changes

[write] Written 5 October 2026 (batch 100).

Sources. The report is built from two manuscripts, delivered as two ZIP archives of the session bundle of Reports 1–243. Both arrived unchanged in ProveIt commit 60f54ea06 (“Arrival: 177 research archives from the session bundle of Reports 1-243”, 5 October 2026) and were placed as cluster DPP of batch 100 in 36571ae0e, which removed the archives from docs/incoming/; they survive in the arrival commit (`git show 60f54ea06:docs/incoming/<archive>`).

	Report 182 (base)	Report 181
archive	Strict_Twice_Partitions_Phase_Asymptotics_Source.zip, 2,961,159 bytes, 38 files	Nested_Partitions_Asymptotics_and_Inverses_Source.zip, 692,760 bytes, 32 files
manuscript title	Report182.tex, 1,148 lines; PDF 30 pages <i>Report182: Phase resolved asymptotics for A271619</i>	Report181.tex, 898 lines; PDF 21 pages <i>Nested Partitions with Distinct Block Lengths: All fixed asymptotic orders, explicit relative growth, inverse bounds and conditioned limits for OEIS A358836</i>
author line, date	“A rigorous asymptotic and reproducibility report” (PDF author “Research report”); 3 October 2026	“Research report 181”; 3 October 2026
pin	none; cites main URLs of the A022629 README and the A291698 article	none; its source audit records the git blob hashes of the three repository files it read (ef1b42ed, 1100837c, 36949d3f), all equal to the files at the write
file prefix printed as	182-strict- Part I, Sections 4–15	181-nested- Part II, Sections 16–27

Neither manuscript says that it is AI-assisted or “prepared for private review”. Neither has been refereed; Report 182 says it has not been submitted externally, and Report 181’s source audit records “No external writes”.

Where the merge chose. Report 182 is the base: A271619 is the sequence that the name “strict twice partitions” denotes, its OEIS entry is the older (2016, against 2022), its mechanism is the more general one (it includes the stretched-amplitude observation), and its text is the longer. Base selection by weaker hypotheses does not apply, since the two manuscripts prove different theorems about different sequences. No result is proved in both, so nothing is printed twice. Both bibliographies are kept in full, Part I’s entries first; the two manuscripts’ entries for the A022629 and A291698 reports ([1] and [31], [2] and [32]) stay separate because each records what its own source inspected. Part II’s OEIS entry is cited under the key `oeisA358836` (the key `oeis` is Part I’s).

Numbering. Sections are numbered continuously: Part I’s Section k is the manuscript’s Section $k - 3$, and Part II’s Section k is Report 181’s Section $k - 15$; theorems, lemmas and propositions are numbered within sections, so they shift with them. Equations are numbered continuously as well: Part I’s equation numbers are Report 182’s, and Part II’s equation (j) is Report 181’s equation ($j - 83$). The delivered READMEs, audits and code comments use the manuscripts’ own numbers. Table 2 maps the numbered results.

Table 2: Numbered results: manuscript numbering and this report’s.

Report	Manuscript	Here
182	Theorem 1.1 (all fixed relative orders)	Theorem 4.1
182	Theorem 1.2 (discrete inverse)	Theorem 4.2
182	Lemma 2.1 (monotonicity)	Lemma 5.1
182	Proposition 2.2 (growth)	Proposition 5.2
182	Theorem 5.1 (tame local expansion)	Theorem 8.1
182	Theorem 6.1 (filled block localization)	Theorem 9.1
182	Lemma 6.2 (quantitative gap)	Lemma 9.2

Report	Manuscript	Here
182	Theorem 9.1 (uniform positive mass approximation)	Theorem 12.1
181	Lemma 2.1 (denominators)	Lemma 17.1
181	Proposition 2.2 (finite end)	Proposition 17.2
181	Theorem 3.1 (free energy)	Theorem 18.1
181	Lemma 4.1 (all-arc bound)	Lemma 19.1
181	Theorem 5.1 (Edgeworth orders)	Theorem 20.1
181	Theorem 6.1 (power-log tiers)	Theorem 21.1
181	Theorem 6.2 (elementary equivalent)	Theorem 21.2
181	Proposition 7.1 (first inverse)	Proposition 22.1
181	Theorem 7.2 (bounded inverse)	Theorem 22.2
181	Theorem 7.3 (two-floor envelopes)	Theorem 22.3
181	Theorem 8.1 (block-count law)	Theorem 23.1
181	Theorem 9.1 (Gumbel law, joint limit)	Theorem 24.1

Editorial changes.

- Labels: Report 182’s 105 labels carry the prefix `stp:sum:`, Report 181’s 81 the prefix `stp:len:`; the write added `stp:sum:part`, `stp:sum:sub:floating`, `stp:sum:sec:further` (the manuscript’s closing section had no label), `stp:len:part`, and the front-matter and appendix labels `stp:sec:guide`, `stp:sec:notation`, `stp:tab:notation`, `stp:sec:limits`, `stp:app:provenance` and `stp:tab:crosswalk`. Every `\ref` and `\eqref` follows its label.
- Preamble: Report 182’s, with Report 181’s macros added; both define $\mathbb{E}$ and Var with the same meaning, and one definition of each is kept. Report 181’s running headers (`fancyhdr`), coloured links and table-of-contents spacing are not carried over; the contents list sections only. The title, author and date lines are new; the PDF metadata names the merged report.
- Abstracts: Report 182’s opens the report, with its finite-index warning and scope paragraph; Report 181’s opens Part II.
- Section titles changed: Report 182’s “Further directions” and Report 181’s “Further questions” are both now “Further questions and research”.
- Citation key: Report 181’s `oeis` is `oeisA358836`.
- Added text: the editorial note, Sections 1–3, the two Part openings, this appendix, and the notes marked [write] in Sections 6, 11, 13, 14, 15, 18, 20, 23, 24, 25, 26 and 27. No other sentence, formula or reference of either manuscript was changed.

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